EC1500V

Refrigeration Cycle Trainer

User Guide

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TecQuipment has taken care to make the contents of this manual accurate and up to date. However, if any errors are found, please let us know so we can rectify the problem.

TecQuipment supply a Packing Contents List (PCL) with the equipment. Carefully check the contents of the package(s) against the list. If any items are missing or damaged, contact TecQuipment or the local agent.

BW/0923

TECQUIPMENT

Symbols Used in this Manual

NOTE		*Important information*
CAUTION		*Failure to carry out this instruction could cause damage to the apparatus, other equipment, personal property, or the environment.*
WARNING		***Failure to carry out this instruction could cause personal injury.***

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EC1500V

Refrigeration Trainer

User Guide

Introduction

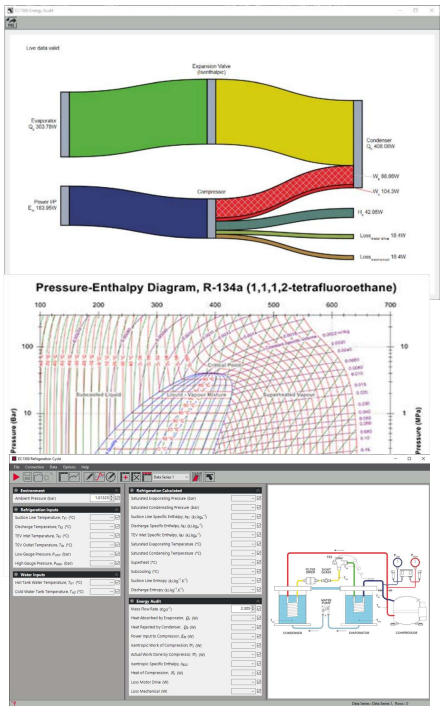




Figure 1 Air Conditioning Trainer, Pressure-Enthalpy Chart, Sankey Diagram and VDAS® Screen

This product works with VDAS® Onboard

Refrigeration cycles are widely applied to many industrial and domestic environments in order to provide the freezing, cooling or heating required. Refrigeration cycles are used in industrial processes for example: chemical processing, pharmaceutical production, food processing and preservation. They are also employed to meet human comfort needs, for example: car or room air conditioning and heating or within a home fridge/freezer.

The term “refrigeration” is defined as cooling a substance to a temperature that is lower than its general environment by moving the heat from one place (heat source) to another place (heat sink). This process is achieved by a device called a “refrigerator”. If the temperature of the heat sink (where the heat is moved to) is high enough for useful applications such as heating water or a room, the thermal device is called a “heat pump”.

The vapour compression refrigeration cycle using a mechanical compressor is the most commonly used refrigeration method for both refrigerating and heating applications. This is due to its high efficiency, the broad range of temperatures it can achieve and its compact size in comparison with other refrigeration methods like vapour absorption or vapour jet cycles.

The Refrigeration Cycle Trainer EC1500V simulates a mechanical refrigeration system. It is capable of demonstrating the fundamental principles of simple vapour compression refrigeration and the main components (necessary for establishing a single-state vapour refrigeration cycle). The student can use the apparatus to investigate the thermodynamic processes of a vapour compression refrigeration or heat pump cycle and determine the cycle's efficiencies along with the effect of different factors on the cycle's performance.

To demonstrate these processes on a Pressure-Enthalpy chart as well as to record experiment results automatically and save time, the equipment has integrated within it TecQuipment's Versatile Data Acquisition System (VDAS®).

Description

EC1500V is built from the individual components required to create a refrigeration circuit that represents a single-stage vapour compression refrigeration cycle.

Key Components:

The vapour compression refrigeration system (VCRS) consists of four basic components that form a simple refrigeration cycle.

- Hermetic-type compressor
- Helical coil condenser
- Thermostatic expansion valve (TEV)
- Helical coil evaporator

The main components of the VCRS are connected to one another by colour-coded refrigerant pipes (Figure 3) and other components including:

- Liquid-line filter drier
- Sight glass
- Service valve (Schrader type)

All the components are mounted together in a compact bench-top frame.

The apparatus has two separate panels. A control panel (on the upper left), comprising the displays, control switches and sockets for instrument connection. Also a refrigeration and water system panel (on the right and lower section), comprising the instruments for the refrigeration system and the components of the water system including two water tanks and a water pump (Figure 2).

Also Included:

- Drip tray
- Laminated wall charts (may be used as posters or for students to use during experiments)
- Digital charts (may be circulated to students to print and use during experiments)

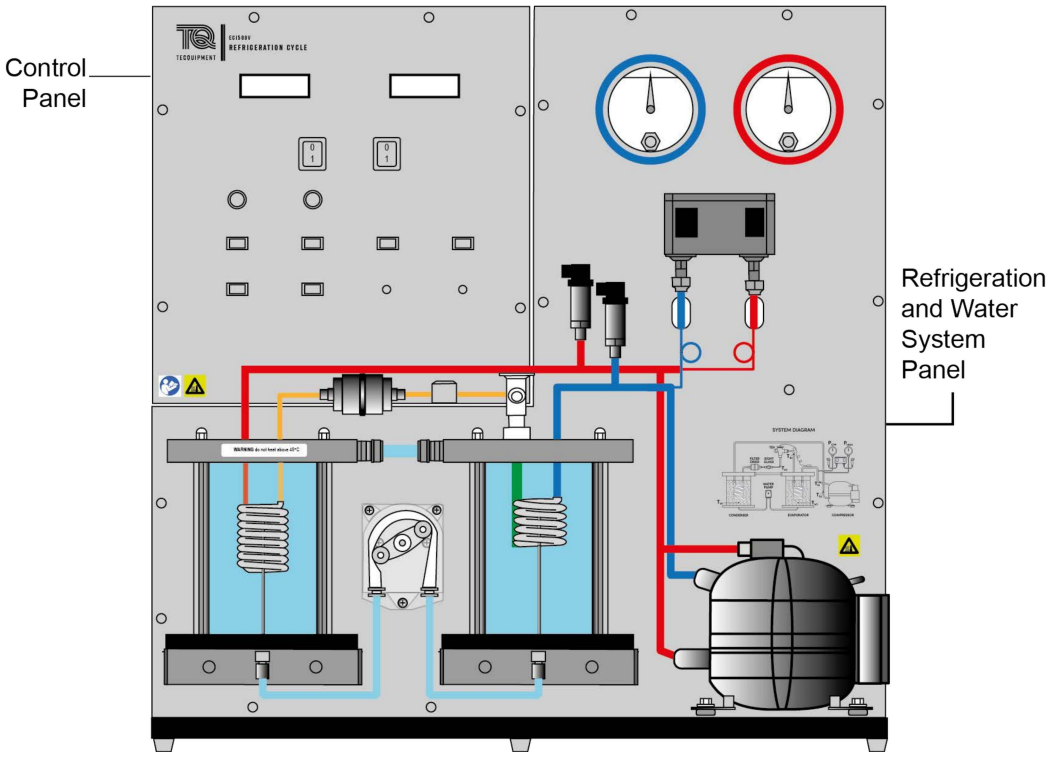


Figure 2 Refrigeration Trainer Apparatus Components

The schematic diagram of the refrigerant circuit with all its components is shown (Figure 3).

The pipes in the schematic diagram are colour coded to represent their contents see Table 1 for key.

Colour	Line Type	Contents
Red		High pressure, high temperature refrigerant vapour
Yellow		High pressure, high temperature refrigerant liquid
Green		Low pressure, low temperature refrigerant mixture (i.e. vapour-liquid)
Blue		Low pressure, low temperature refrigerant vapour

Table 1 Colour Coding of Pipes

Colour	Line Type	Contents
Pale Blue		Water

Table 2 Other Colours Used on the Schematic Diagram

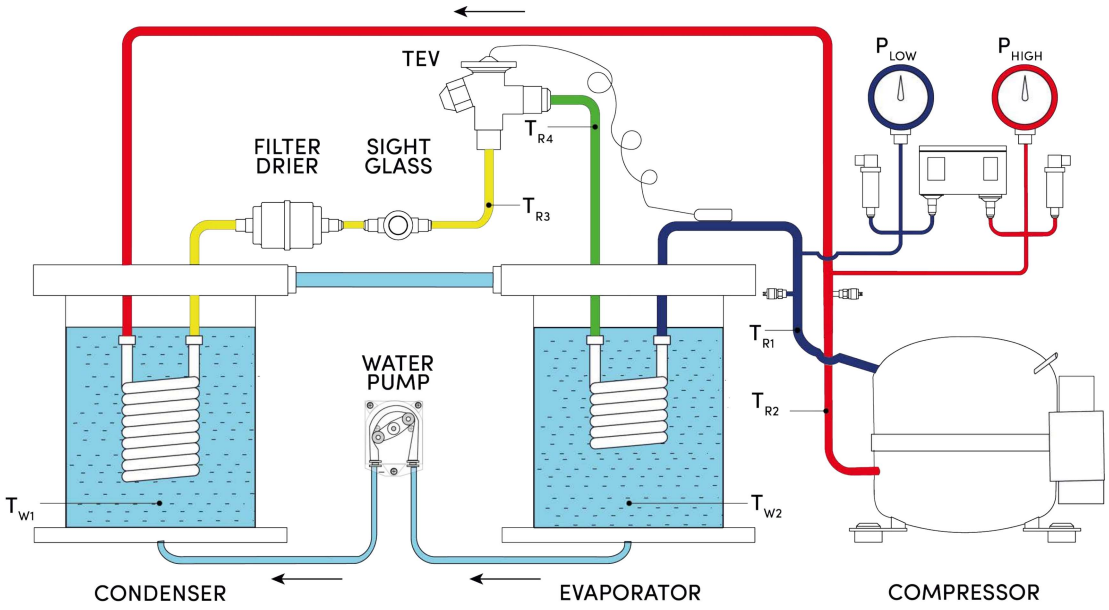


Figure 3 Refrigeration System Schematic

To correlate with the instrument labels, the symbols for the measurements are as follows:

Symbol	What it means or connects to
T_{R1}	Suction-line temperature
T_{R2}	Discharge-line temperature
T_{R3}	Liquid-line temperature
T_{R4}	Liquid/vapour mixture-line temperature
P_{HIGH}	High-side pressure
P_{LOW}	Low-side pressure
T_{W1}	Hot water tank temperature (i.e. heat sink temperature)
T_{W2}	Cold water tank temperature (i.e. heat source temperature)

Table 3 Variables Measured by the System

The apparatus contains R134a as a refrigerant. It is known as HFC-134a with a chemical name of 1,1,1,2-tetrafluoroethane. This is a refrigerant that is non-flammable, non-corrosive, has low toxicity and good thermal stability. It is commonly used in automotive air conditioning and commercial refrigeration systems.

CAUTION  *The refrigeration system is supplied charged with the correct amount of refrigerant. In the case of a suspected refrigerant leak, please contact a certified service engineer to fix the problem before attempting to carry out any experiments.*

Refrigeration and Water System Panel

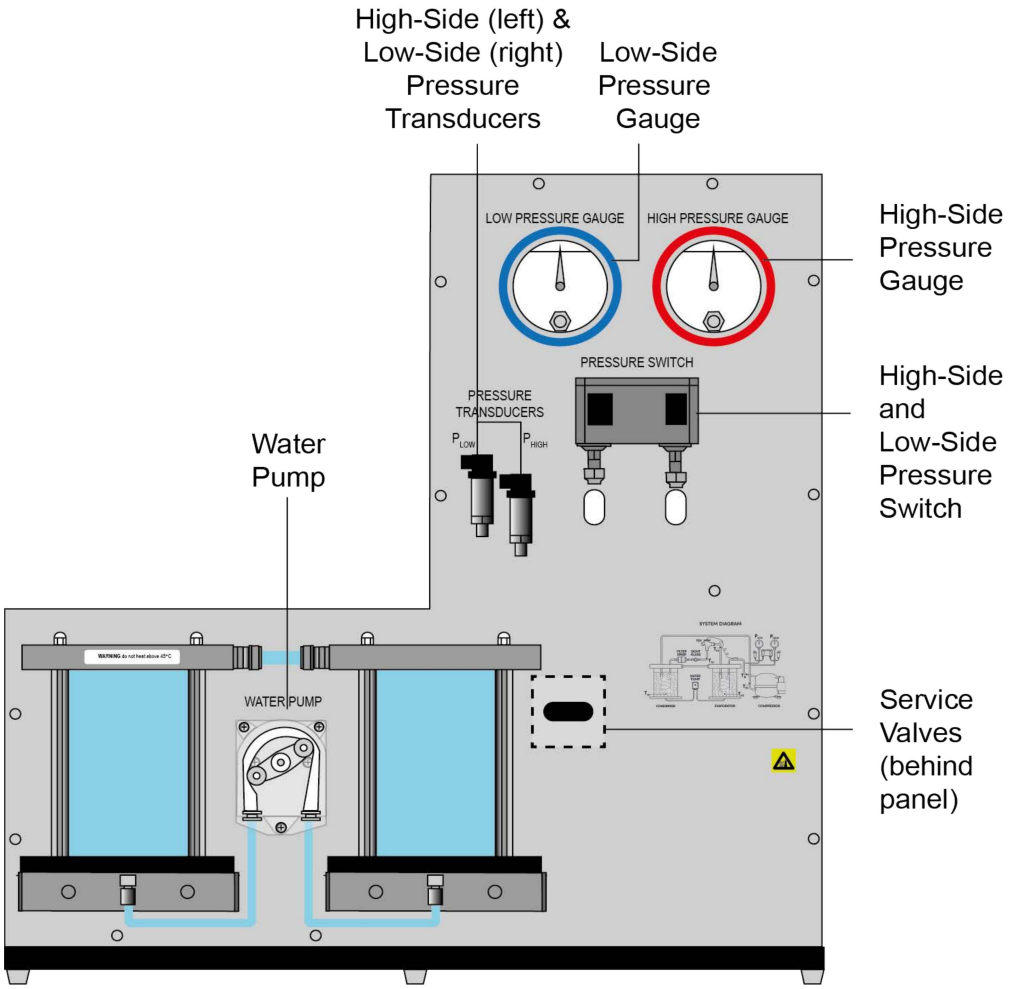


Figure 4 Refrigeration Instrument Panel

The components of the Refrigeration Instrument Panel are shown above (Figure 4).

Water Tanks and Pump

The left water tank houses the condenser coil, this represents the heat sink. The right water tank houses the evaporator coil, this represents the heat source.

Both the evaporator and condenser coils are submerged in the water tanks. A small pump provides a gentle mixing of the water between the two tanks to help maintain a steady state condition. The water is pumped from the right tank (cold tank) to the left tank (hot tank).

First Stage Protection Device

The system is protected from pressure becoming too high or too low in the circuit by an automatically resetting pressure switch located on the instrument panel. See **Figure 6** on page 8.

The pressure of the low-side is shown on the blue gauge on the refrigerant panel. If the pressure becomes too low the switch will trip and the system will shut down. The red low pressure warning lamp on the control panel will illuminate to indicate that there is a problem with the low pressure part of the system.

The pressure of the high-side is shown on the red gauge on the refrigerant panel. If the pressure becomes too high the switch will trip and the system will shut down. The red high pressure warning lamp on the control panel will illuminate to indicate that there is a problem with the high pressure part of the system.

Resetting the System after a Pressure Warning

WARNING 

The pressure switch is factory set at default values as follows:

Low pressure side: Pressure 1 bar, Differential 1 bar

High pressure side: Pressure 14 bar, Differential 4 bar fixed.

Changing these default settings may result in potential risks and the incorrect operation of the apparatus.

WARNING 

If either of the pressure warning lamps are illuminated, or the unit is damaged in any way, do not attempt to repair or re-start it. Please isolate the apparatus and contact TecQuipment or local agent.

Second Stage Protection Device

In the unlikely event that the first stage protection device has a fault, the system continues to be protected from over-pressure on the high pressure side by an electro-mechanical pressure switch fitted in the compressor's discharge line.



Figure 5 Second Stage Protection Device

This switch will trip to cut off the compressor at approximately 18 bar (250 ± 15 PSI) and cut the compressor in at about 10 bar (150 ± 15) PSI.

Control Panel

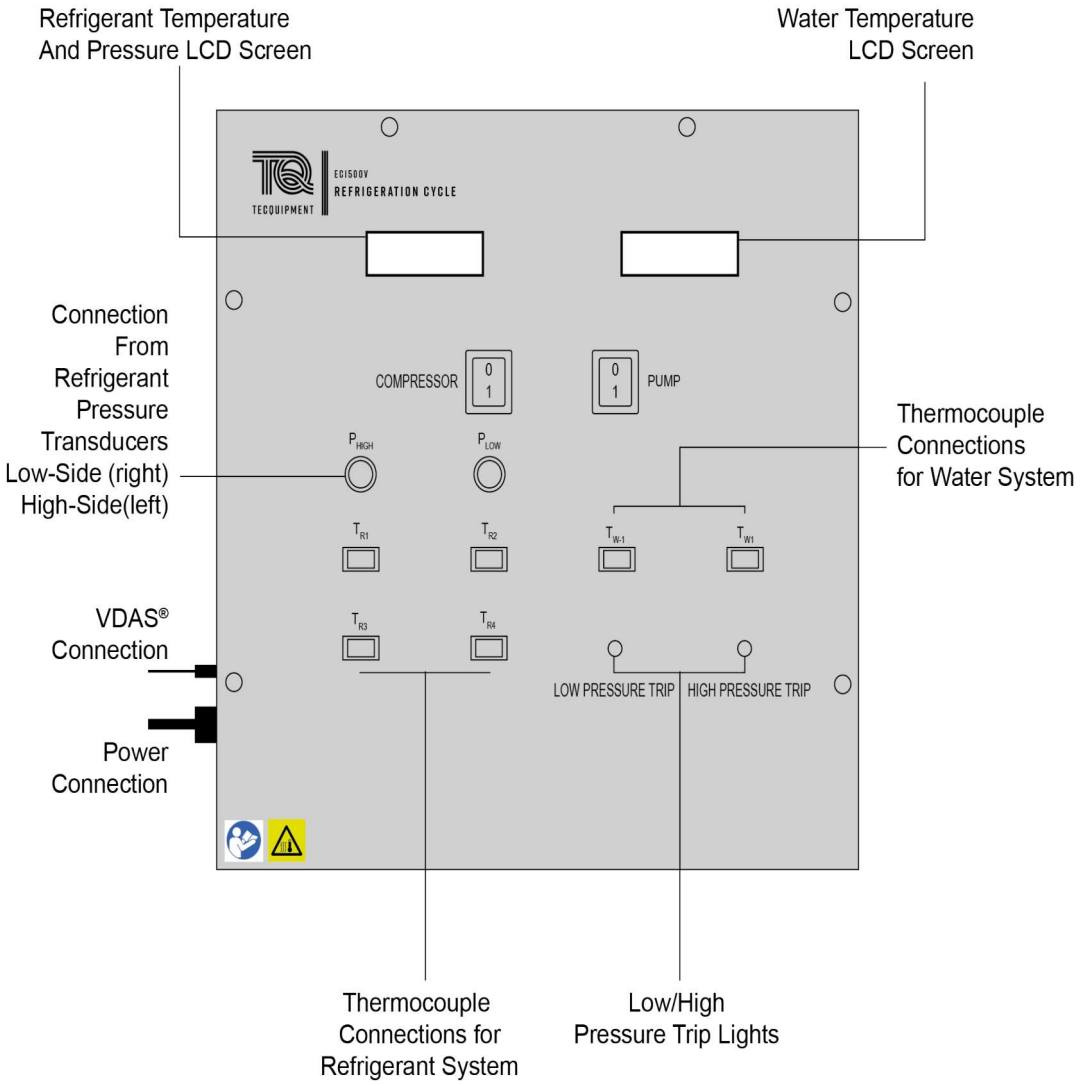


Figure 6 Control Panel

The control panel is located at the back of the unit on the left. The main isolator for the apparatus, power and VDAS® connections are located on its left hand edge. The switches for the compressor and water pump are located on its front face. Also on the front face are the LCD display screens for refrigerant temperature and pressure (left) and water temperature (right) (Figure 6).

The control panel has the connection points for the four thermocouples measuring refrigerant temperatures and two thermocouples measuring water temperatures. It also houses the sockets for the refrigerant high and low pressure sensors.

There are two warning lamps on the panel, when illuminated these indicate if the system has been tripped off by either a low-side or high-side pressure warning (one lamp for each). In the event that the system trips due to a low or high faults see **Resetting the System after a Pressure Warning** on page 7.

Measurements

The temperatures and pressures of the refrigerant around the refrigeration circuit are measured by means of two pressure transducers that are fitted on both the high and low pressure sides, and four type K thermocouples that are attached on the surface of the refrigerant pipe around the circuit. See **Figure 3** on page 5.

Two pressure gauges are also connected to the refrigeration circuit. These show the pressure and saturation temperatures of the refrigerant. These may be used for calculation, but greater accuracy is achieved by the pressure transducers and using VDAS® to calculate the saturated temperatures.

The water temperatures of the two water tanks are measured by means of two stainless steel Type K thermocouples placed at the centre of the water tanks.

These measurements are all displayed on the LCD display screens on the front panel (refrigerant on the left and water on the right).

Versatile Data Acquisition System (VDAS®)

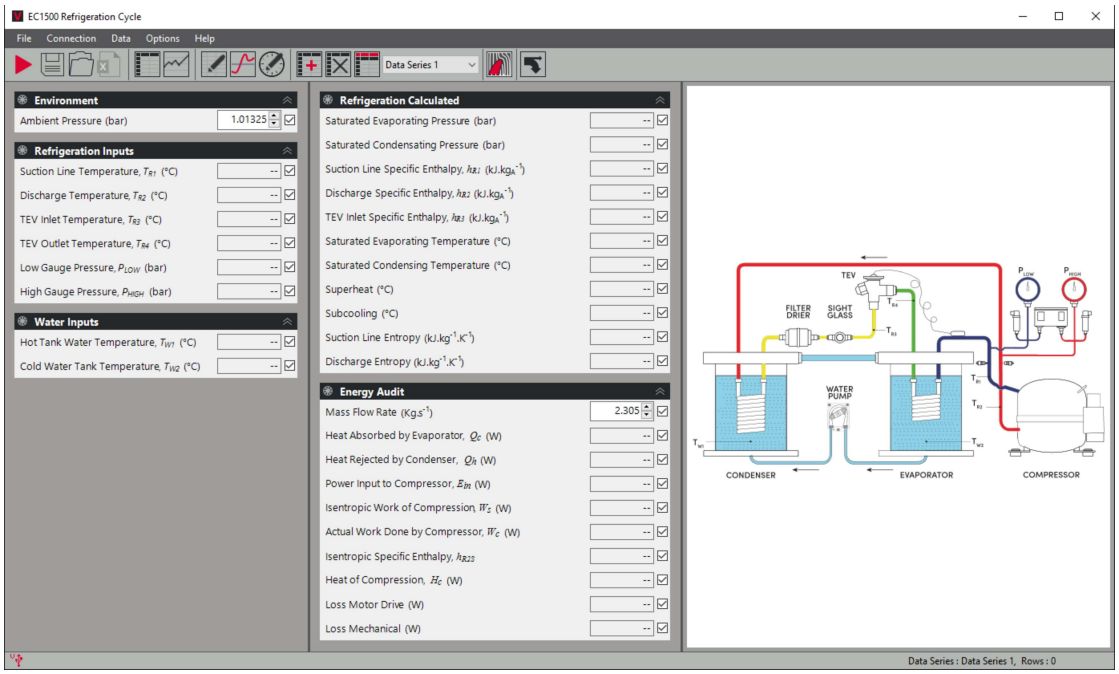


Figure 7 The VDAS® software

The EC1500V includes integrated data acquisition hardware that allows the unit to be directly connected to a computer via a USB cable (provided with the unit) this connects on the left hand edge of the unit control panel. No additional hardware is required. Our VDAS® software is fully compatible and provides the following features:

- Automatically logs data from experiments
- Automatically calculates experiment parameters
- Saves time
- Reduces errors
- Creates charts and tables of data
- Exports data for processing in other software
- Includes Pressure-Enthalpy chart
- Includes Sankey Diagrams

NOTE *A suitable computer is needed (not supplied) to use TecQuipment's VDAS®.*

Operation of VDAS®

A separate manual is supplied for the VDAS® software.

The EC1500V VDAS® software screen shows all measured data and calculated results together with the refrigeration cycle on a Pressure-Enthalpy Chart and Sankey Diagrams. This is functionality that is not usually available to other VDAS® compatible products. Please see below for how to access this additional functionality.

Pressure-Enthalpy Chart

The Pressure-Enthalpy chart icon (Figure 8) is located in the toolbar (Figure 9) at the top of the main VDAS® screen, open the chart window. The chart overlays a 'live' view of the current measurement data, when connected to the EC1500V apparatus.

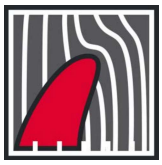


Figure 8 Pressure-Enthalpy Chart Icon

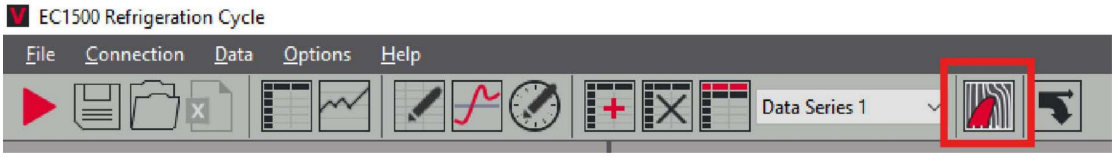


Figure 9 VDAS® Toolbar showing Pressure-Enthalpy Icon

Saving a Digital Copy of a Pressure-Enthalpy Chart

In order to save a digital copy of a Pressure-Enthalpy chart, right click anywhere on the chart while it is on screen (Figure 11). The following selection box will appear at the cursor (Figure 10):

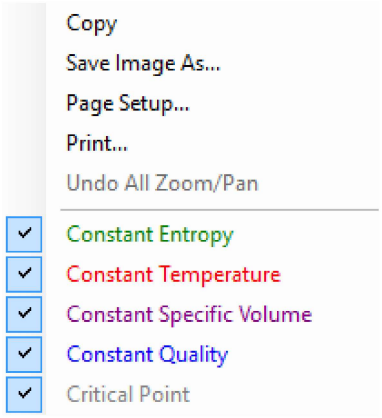


Figure 10 Pressure-Enthalpy Chart Selection Box

Select ‘Save Image As’ and follow the on screen instructions to name and save your file for importing into other applications e.g. word processed reports.

Change the Pressure-Enthalpy Chart Display

It is possible to ‘switch off’ any of the sets of lines displayed on the Pressure-Enthalpy chart using the same selection box. Again right click on the chart to see the box at the cursor (Figure 11).

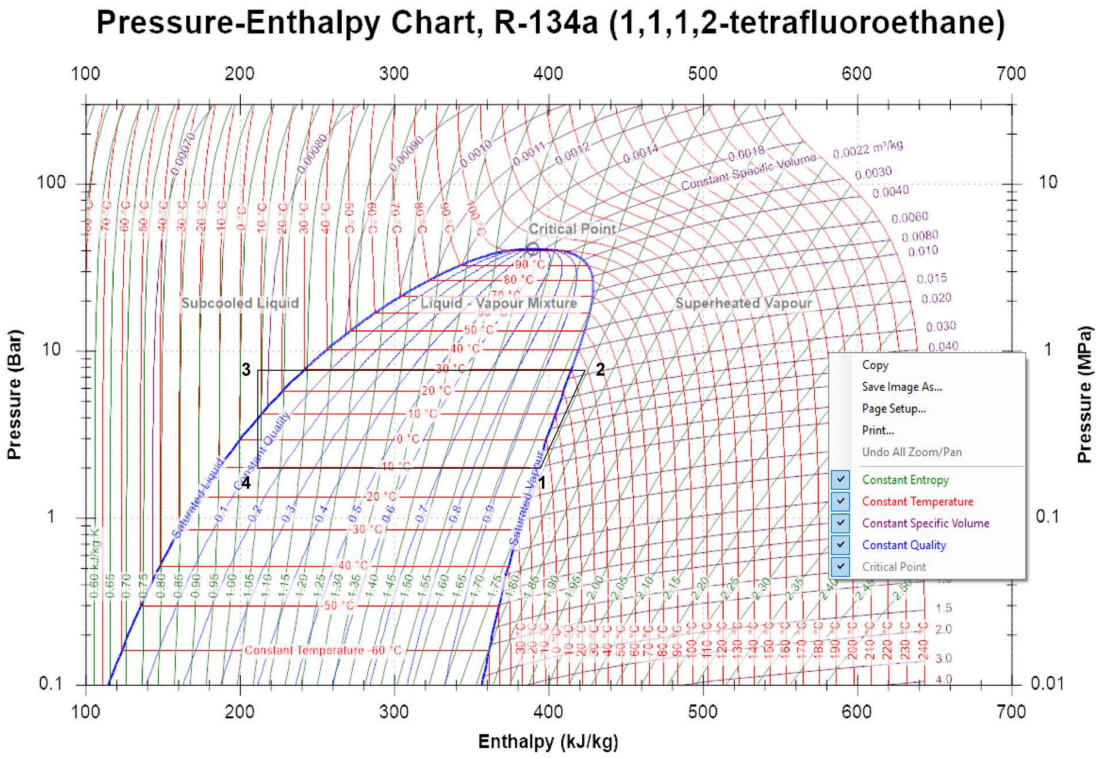


Figure 11 P-h Chart Showing Selection Box

Check or uncheck any combination of the five boxes labelled:

- Constant Entropy
- Constant Temperature
- Constant Specific Volume
- Constant Quality
- Critical Point

To display or otherwise sets of lines on the chart.

NOTE *The colour of the text of the labels in the selection box corresponds to the colour of the sets of lines on the chart.*

Figure 12 shows a chart where both Constant Entropy and Constant Temperature have been unchecked. It can be seen that both the Entropy (green) lines and Temperature (red) lines have been switched off. This has resulted in the Specific Volume (purple) and Quality (blue) lines appearing much clearer also the Critical Point.

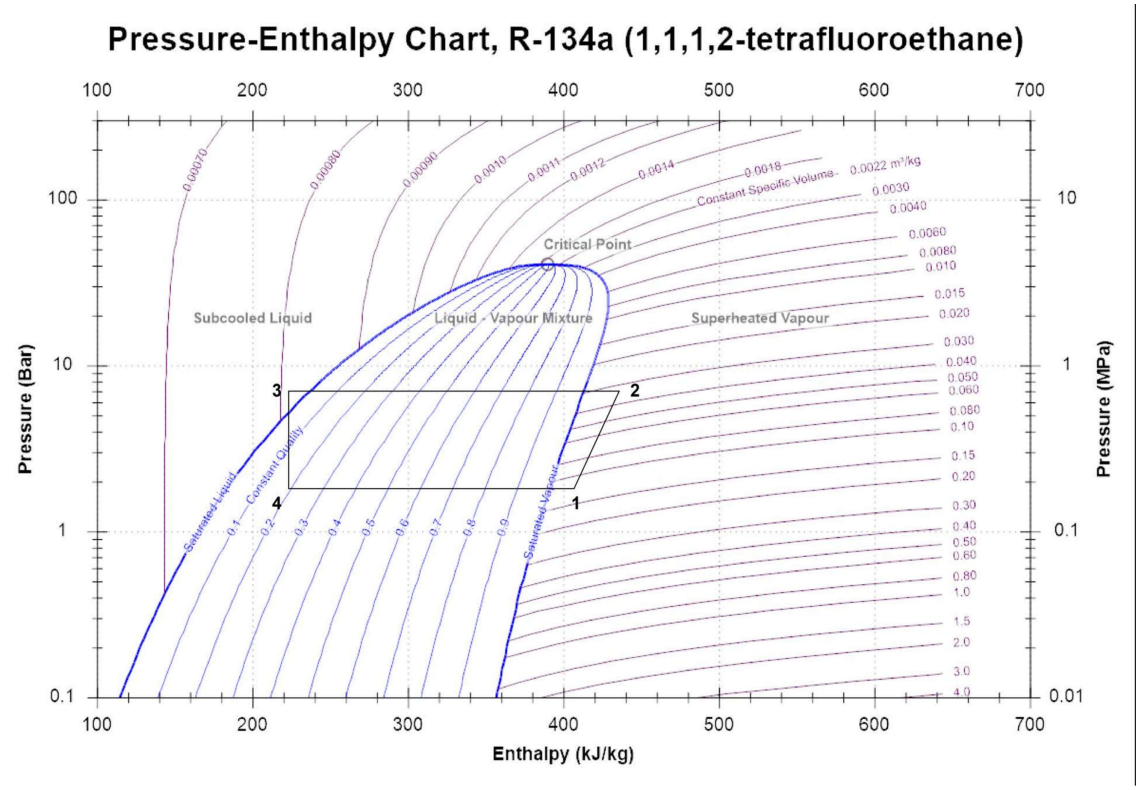


Figure 12 P-h Chart showing only Specific Volume and Quality Lines along with the Critical Point

Sankey Diagram

A Sankey diagram is a visualisation used to depict a flow from one set of values to another. Clicking on the EC1500V Sankey diagrams icon shows a snapshot of the flow of energy through the apparatus.



Figure 13 Sankey Diagram Icon

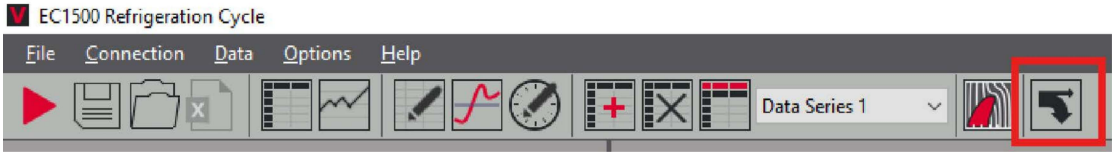


Figure 14 VDAS® Toolbar showing the Sankey Diagram Icon

An example Sankey diagram can be seen in (Figure 15).

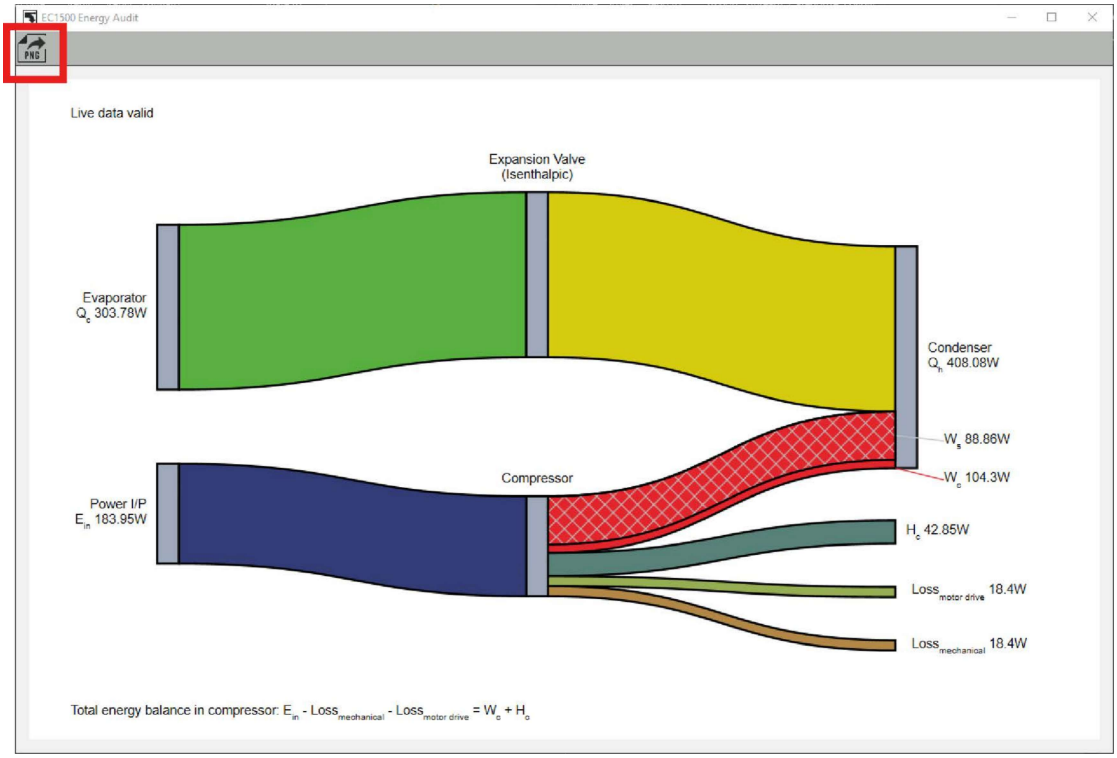


Figure 15 Example Energy Flow Sankey Diagram (showing the export to png icon)

A png version of a Sankey diagram can be captured by selecting the png Icon on while viewing a diagram (Figure 15) and (Figure 16).



Figure 16 Export Sankey Diagram to PNG Icon

Technical Details

Main Frame

Item	Details
Nett dimensions	825mm (L), 494mm, (D) 845mm (H), and mass 58kg (approx).
Electrical supply (specified on order)	Single-phase; Single phase, 220 - 240 VAC, 50 Hz, 2A **OR** Single phase, 110 - 120 VAC, 60 Hz, 4A **OR** Single phase, 208 - 220 VAC, 60 Hz, 2A
Operating environment	Laboratory Storage temperature range: –25°C to +55°C (when packed for transport) Operating temperature range: +5°C to + 30°C
External connections	VDAS® socket on the left side of the Control Panel
Module pressure trips	Low pressure side: Pressure 1 bar, Differential 1 bar High pressure side: Pressure 14 bar, Differential 4 bar

Table 4 Specification

Main Components	Type	Specifications
Compressor	Hermetic reciprocating	1/8 Horsepower, displacement of 4 cm^3, refrigeration capacity of 329 W rating at the standard condition as rated in EN 12900
Condenser	Helical coil	12 turns of 1/4” OD copper tube immersed in a water tank, heat transfer surface area of about 0.082 m^2
Evaporator	Helical coil	7 turns of 1/4” OD copper tube immersed in a water tank, heat transfer surface area of about 0.049 m^2
Expansion device	Thermostatic expansion valve	Internally equalized type
Refrigerant	R134a	Hydrofluorocarbon (HFC), boiling temperature of -26.4^{o}C at the atmosphere pressure of 101.3 kPa

Table 5 Technical specifications of the main components used in EC1500V

Refrigerant Details

The equipment contains fluorinated greenhouse gas.

Item		Units
Type of F-Gas	HFC-134a (1,1,1,2-Tetrafluoroethane)	-
Total F-Gas mass	This information is unit specific. Please refer to the label on each unit for the specific information.	kg
Total eq (C_2)		tonnes
GWP	1430	-
Maximum allowable pressure	14	bar

Noise Levels

The Noise Levels Recorded at this Apparatus	Volume
Within 20 cm of the water pump	80 dB
40 cm away from the water pump	Less than 70 dB

Assembly and Installation

The terms **left**, **right, front** and **rear** of the apparatus refer to the operators' position, facing the unit.

NOTE *Obey any regulations that affect the installation, operation and maintenance of this apparatus in the country where it is to be used.*

Location and Assembly

The EC1500V is supplied ready assembled.

Lifting and Carrying

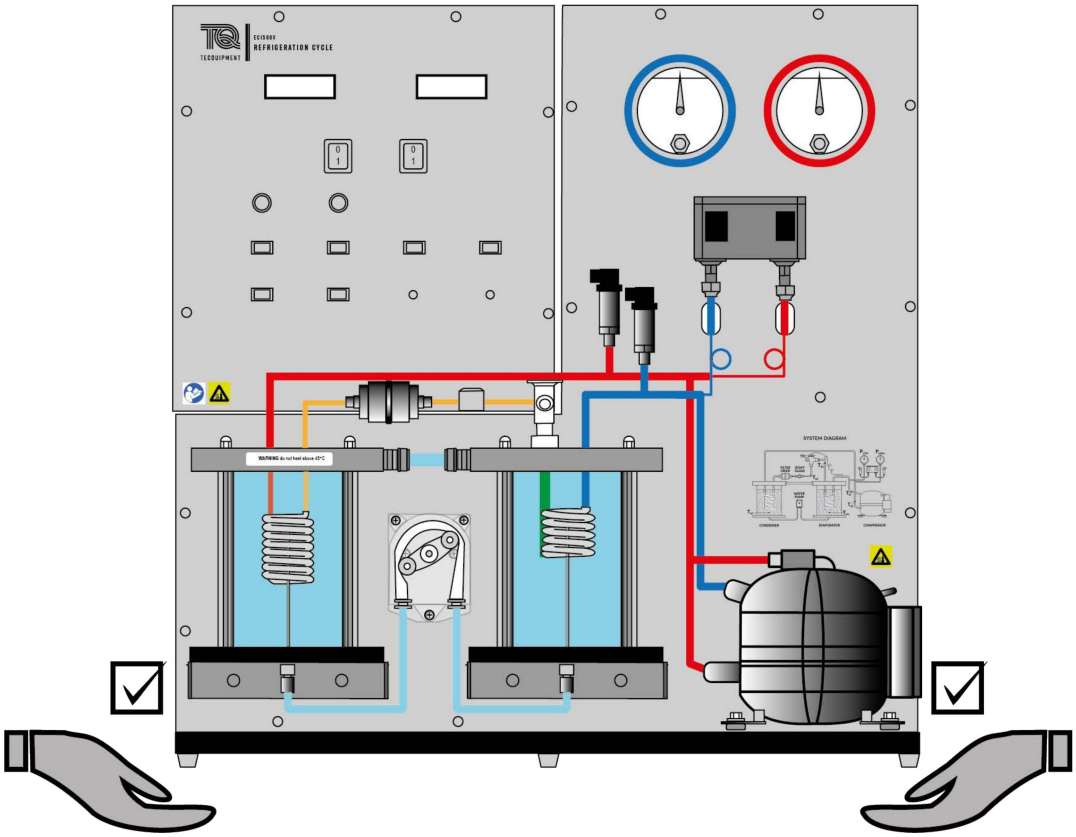


Figure 17 Safe Lifting Points EC1500V

WARNING ***The EC1500V is heavy (see Technical Details). Use assistance and suitable lifting equipment to lift, carry or move it around.***

Use the equipment in a clean, well-lit laboratory or classroom type area. Use assistance to put it on top of a solid, level workbench for safe lifting points (see Figure 17 page 19).

The equipment requires a minimum bench area of roughly 850 mm x 500 mm. If using the VDAS® data capture, allow room nearby for a computer.

Equipment set-up

Refrigerant Thermocouple Connection

Four type K thermocouples together with their cables and plugs (T_{R1}, T_{R2}, T_{R3}, T_{R4}) are supplied. They must be connected to the control box via sockets on the control panel.

Please check the labels on each connector and plug the cables into appropriate sockets See **Figure 18** on page 21.

Water Thermocouple Connection

Two stainless steel type K thermocouples together with their cables and plugs (T_{W1} and T_{W2}) are supplied. They must be connected to the control box via sockets on the control panel.

Please check the labels on each connector and plug the cables into the corresponding sockets See **Figure 18** on page 21.

Refrigerant Pressure Transducer Connection

Two pressure transducer sensors (high and low pressure transducers) together with their cable/connector must be connected to the control box via sockets on the control panel. Please refer to the labels on the front panel for the appropriate connections (the high pressure transducer connects to the P_{HIGH} socket and the low pressure transducer connects to the P_{LOW} socket). For the location of the connection points See **Figure 18** on page 21.

Drip Tray

Place the drip tray under the right tank right at the back with its right side against the red drip shield in order to catch the condensed water that forms on the low-side pressure pipe (blue).

Refrigeration Instruments Rear Cover

The metal cover over the rear of the instrument panel instruments may be removed for teaching purposes if required. However, it is recommended that it is replaced as soon as possible and kept in place to protect fragile parts from damage.

Filling with Water

The water tanks are located on the front of the unit, they do not have lids. Both should be filled via their open top with clean water (preferably de-mineralised or de-ionised, at about 20°C to 23°C) to the base of the tank's top plate (Figure 18).

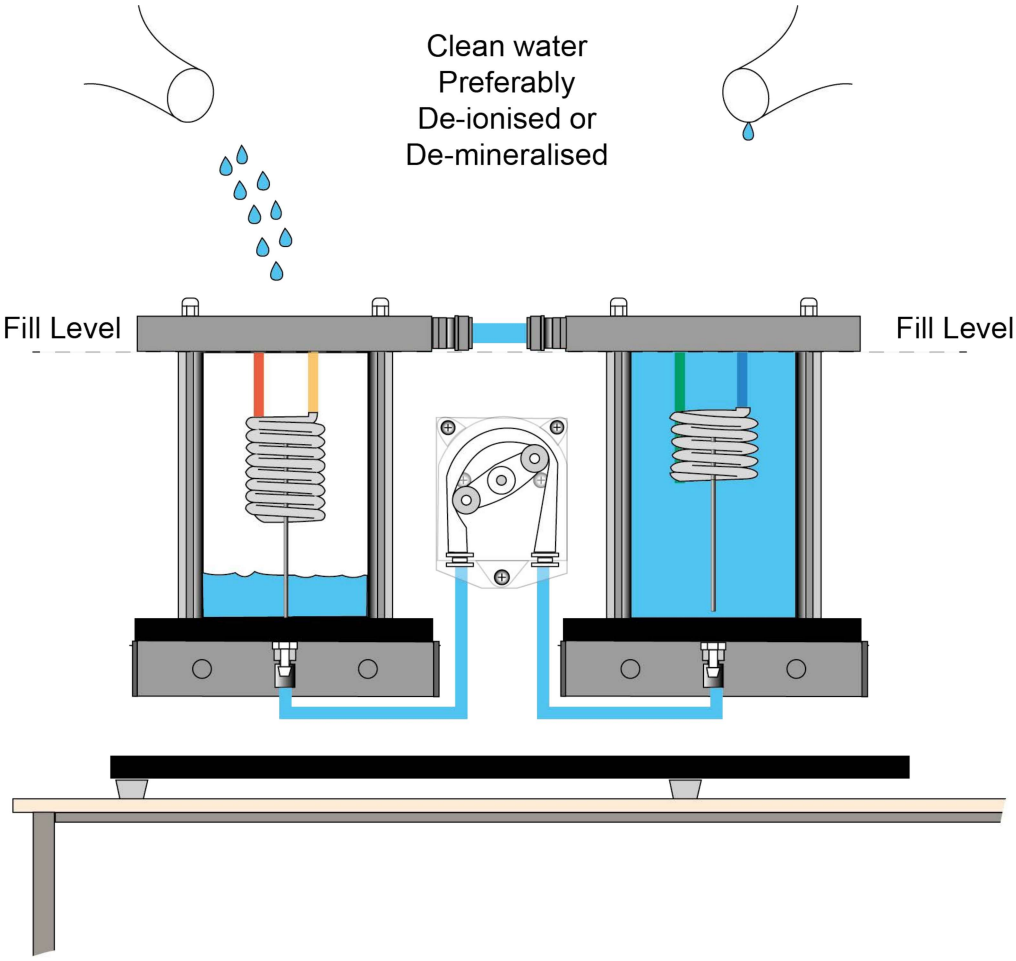


Figure 18 Fill level

WARNING ***Take care not to splash the water onto the equipment. Always clear up water spills immediately.***

NOTE *Use de-ionised or de-mineralised water where possible*

Connection to VDAS®

Unscrew and open the dust cap covering the VDAS® connection point on the left hand edge of the control panel and fit the supplied USB cable to connect to a suitable computer (Figure 19).

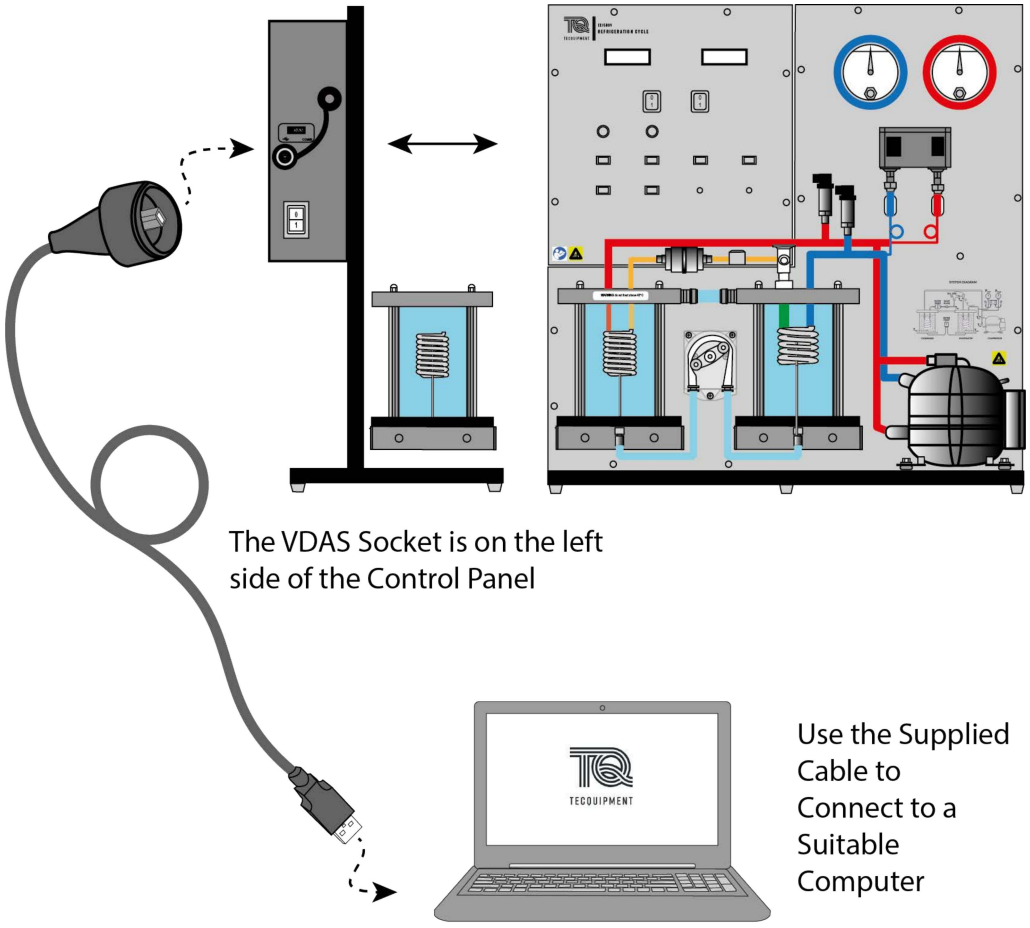


Figure 19 VDAS Connection

CAUTION *Always replace the dust cap to the VDAS® socket when not in use to prevent damage from dust or water entering.*

Electrical Connection

WARNING  ***Connect the equipment to the electrical supply through a plug and socket. The apparatus must be connected to earth.***

Use the cable supplied to connect the equipment to the electrical supply.

Then switch on the main isolator located on the left side of the control panel.

These are the colours of each individual conductor:

Single Phase Supply:

E / Earth / Ground / ⏚	**Green / Yellow**	**OR**	**Green**
L / Live / Phase / Hot	**Brown**	**OR**	**Black**
N / Neutral / Cold	**Blue**	**OR**	**White**

Start Up Procedure

WARNING  ***If someone has not used the equipment correctly and it is suspected that water has accidentally entered the control panels, disconnect the electrical supply and ask a qualified electrician to check the equipment before using it again.***

CAUTION  *Using dirty water can cause problems. Any minerals or dirt in the water will slowly affect the accuracy of the sensors and heat treansfer capability of the helicoil coils (preferably use de-mineralised or de-ionised water).*

NOTE  *Only use the supplied USB cable as it has a water seal inside. When not in use please re-fit the connector cap (Figure 19).*

NOTE  *Obey any regulations that affect the installation, operation and maintenance of this apparatus in the country where it is to be used.*

Once the unit has been connected to the mains supply:

1. Fill the two water tanks to the base of the top plate with clean water of about 20°C to 23°C (preferably de-mineralised or de-ionised) (Figure 18).
2. Plug the unit into the mains power supply.
3. Switch ON the main isolator switch (located on the left hand edge of the control panel).
4. Switch ON the pump (the switch is located on the front of the control panel). Depending on the experiment being performed, the pump may not be required.

NOTE  *The water will rise a little in the left (hot) tank and drop in the right (cold) tank. Top up the right tank so it is once again just below the bottom of the top plate (Figure 18).*

5. Leave the unit to settle until the LCD readings stabilise before it is ready to use. Depending on local conditions, this could take 30 minutes or more.

CAUTION  *Always leave the compressor for at least five minutes after it has been switched off before it is switched on again, or it could be damaged.*

6. Switch ON the compressor when instructed to do so for the experiment being performed (the switch is located on the front of the control panel).

Shut Down Procedure

1. Switch OFF the compressor (the switch is located on the front of the control panel).

CAUTION  *Always leave the compressor for at least five minutes after it has been switched off before it is switched on again, or it could be damaged.*

2. Switch OFF the pump (the switch is located on the front of the control panel).
3. Switch OFF the main isolator (located on the right hand edge of the control panel).
4. Disconnect the power supply.
5. Disconnect the VDAS® cable and replace the cover over the socket. See **Figure 19** on page 22.
6. If the unit is not going to be used again in the immediate future drain the water tanks. See **Draining the Water** on page 25.
7. Leave the unit to settle until condensation stops forming (this will take 45 minutes or more depending on local conditions), before drying off any moisture and emptying drip tray.

NOTE  *The water must be changed at least once a week.*

8. If necessary, clean the tanks See **Removal of Tanks for Cleaning** on page 27.

Draining the Water

1. Ensure that there is a suitable container below the apparatus (Figure 21)
2. Put one end of the hose into the container, connect the other end of the hose to the drainage valve on the bottom of the right tank, gently push the hose into the hole on the bottom of the valve until it clicks (the valve opens automatically).

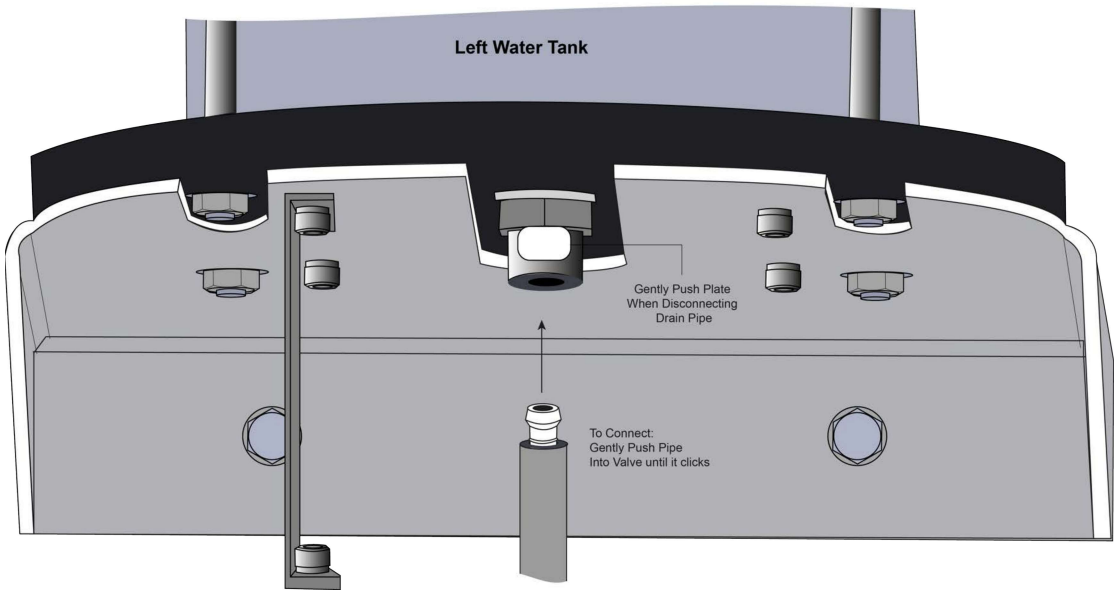


Figure 20 Drain Valve Under Tank

3. Drain the left tank once the right one has drained by disconnecting the hose from the right tank and fitting it to the left one as in step 2 above (make sure that the other end of the hose remains in the container).

4. After draining both tanks, switch the pump ON to pump any remaining water from the pipes. (leave the hose connected to the left tank). Keep the pump running until no more water is entering the left tank.

5. Once the water has drained off, disconnect the drain tube by pressing the plate on the front of the valve (the valve will close automatically) while gently pulling the pipe (Figure 20).

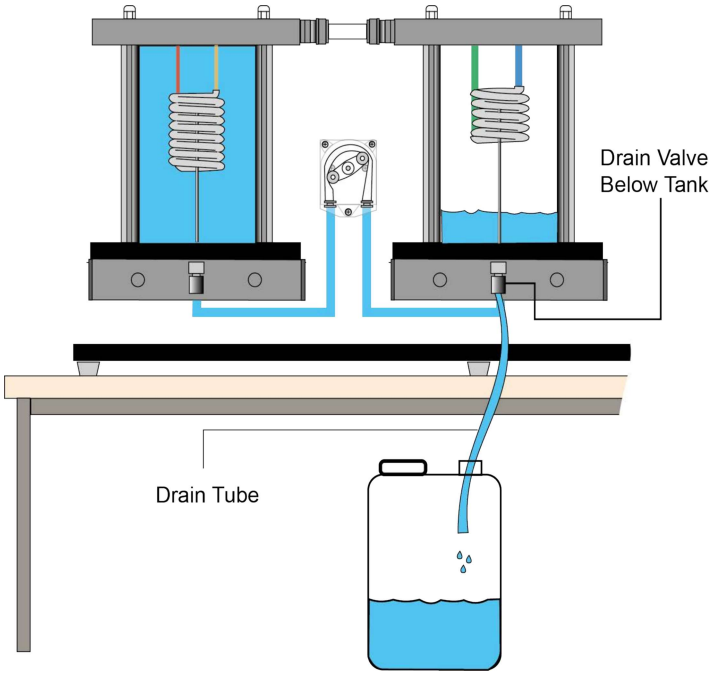


Figure 21 Draining the tanks

Removal of Tanks for Cleaning

CAUTION *Take great care not to damage the refrigeration pipework when removing the tanks.*

1. Undo the nut (1) connecting the water circulation pipe to the top rim of the tank (Figure 22).
2. Unscrew the two screws (2) connecting the bracket that supports the tank base plate (Figure 23).
3. Remove the bracket (A) in the direction shown (Figure 23).
4. Unscrew the three screws (3) connecting the tank base plate to the tank base (Figure 23).
5. Undo the two screws and nuts (4) connecting the tank base plate to the equipment front panel (Figure 23).
6. Whilst carefully holding the tank, remove the tank base plate (B) in the direction shown (Figure 23).
7. Move the tank in direction (C) so that it is clear of pipe (4) (Figure 22).
8. Lower the tank and carefully take it clear of the coil in direction (D) (Figure 22). Ensure the thermocouple clears the coil when rotating the tank out.
9. Repeat steps 1 to 8 for the right tank. adjusting the directions parts are moved in accordingly.

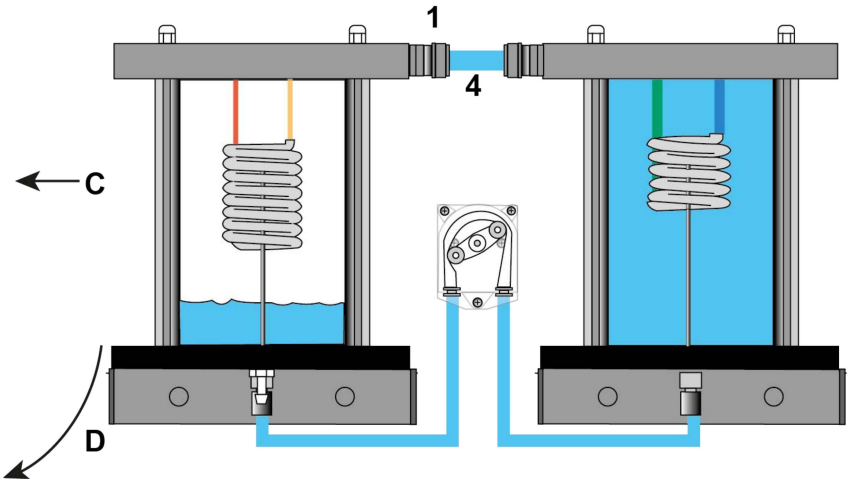


Figure 22 Removing the Tank (1)



Figure 23 Left Tank Base Plate

Reinstall Tank

To reinstall the tanks follow the removal directions above in reverse order.

Notation, Useful Equations and Theory

This section only gives the basic information needed to do the experiments. For full theory, refer to the useful textbooks section. See **Useful Textbooks** on page 77.

Notation

Symbol	Definition	Units
C	Percentage clearance of compressor	%
COP	Coefficient of performance	
COP_c	Cooling coefficient of performance	
COP_{c_carnot}	Cooling coefficient of performance of Carnot cycle	
COP_h	Heating coefficient of performance	
COP_{h_carnot}	Heating coefficient of performance of Carnot cycle	
DSH	Discharge superheat	K
h, h_{R1}, h_{R2}, h_{R3}, h_{R4},	Specific enthalpy of refrigerant at compressor suction, compressor discharge, TEV inlet, TEV outlet	$kJ.kg^{-1}$
h_{R2s}	Specific enthalpy of refrigerant at compressor discharge under isentropic compression	$kJ.kg^{-1}$
n	Polytropic exponent	
p	Pressure	kPa
P_{atm}	Atmospheric pressure	kPa
P_{GAUGE}	Gauge pressure	bar
P_{LOW}, P_{HIGH}	Gauge pressures of low-side and high-side	bar
P_{LOW_ABS}, P_{HIGH_ABS}	Absolute pressures of low-side and high-side	bar
P_{ratio}	Pressure ratio of compressor	
P_{r1}, P_{r2}	Reduced pressures under suction and discharge pressure condition	kPa
q	Heat transfer rate	$kJ.kg^{-1}$
q_c	Refrigerating effect	$kJ.kg^{-1}$
q_{c_carnot}	Refrigerating effect of Carnot cycle	$kJ.kg^{-1}$
q_h	Heating effect	$kJ.kg^{-1}$
q_{h_carnot}	Heating effect of Carnot cycle	$kJ.kg^{-1}$
r	Latent heat of vaporisation or condensation	$kJ.kg^{-1}$
s_A, s_B, s_C, s_D	Specific entropy of initial and final states of isentropic and isothermal processes in Carnot cycle	K
SC	Subcooling degree	K
SH	Superheat degree	K

Symbol	Definition	Units
T_A, T_B, T_C, T_D	Absolute temperatures of initial and final states of isentropic and isothermal processes in Carnot cycle	K
T_c	Critical temperature of refrigerant	K
T_{GAUGE}	Pressure gauge based temperature	°C
T_{R1}, T_{R2}, T_{R3}, T_{R4}	Refrigerant temperature, at compressor suction, compressor discharge, TEV inlet, TEV outlet	°C
T_{r1}, T_{r2}	Reduced temperature under suction and discharge temperature condition	°C
T_{sat}	Saturated temperature of refrigerant	°C
$T_{sat_condensation}$	Saturated temperature of refrigerant at condensing pressure	°C
$T_{sat_discharge}$	Saturated temperature of refrigerant at discharge pressure	°C
$T_{sat_suction}$	Saturated temperature of refrigerant at suction pressure	°C
T_{W1}, T_{W2}	Hot water tank (heat sink) and cold water tank (heat source) temperatures	°C
u	Specific internal energy	$kJ.kg^{-1}$
V	Volume	m^3
v	Specific volume	$m^3.kg^{-1}$
$v_{suction}$	Specific volume of refrigerant at compressor suction	$m^3.kg^{-1}$
w_c	Compression work of actual cycle	
w_{carnot}	Compression work of Carnot cycle	$kJ.kg^{-1}$
w, w_s	Compressor work of actual and isentropic compression processes of refrigeration cycle	$kJ.kg^{-1}$
x	Refrigerant quality (dryness)	kPa
Z_1, Z_2	Compressibility factors under suction and discharge pressure conditions	
η_s	Isentropic efficiency	%
η_v	Volumetric efficiency	%

Table 6 Symbols and Nomenclature

The Principles of Refrigeration

A refrigeration system is a combination of different components and devices connected sequentially. Refrigerant passes in a closed loop through the components between a heat source where heat is absorbed causing a refrigerating effect in that area, to a heat sink where the absorbed heat is released resulting in a warming of that area.

Refrigeration utilises the nature of the latent heat of vaporisation: the absorbing of heat from a heat source when a liquid refrigerant is vaporised. Similarly, the latent heat of condensation is discharged when the refrigerant vapour is liquefied, resulting in raising the temperature in the heat sink.

Latent heat of Vaporisation and Condensation

Latent heat is the heat that is absorbed or released by a substance at a constant temperature resulting in a change of its physical state (phase change). There are three basic types of latent heat associated with phase changes.

	Type of Latent Heat	Endothermic Phase Changes	Exothermic Phase Changes	Temperature
Solid to Liquid	Fusion	Melting	Solidification, freezing, crstallisation	Melting or freezing
Liquid to Vapour[a]	**Vaporisation**	**Boiling, evaporation**	**Condensation, liquification**	**Boiling point or dew point**
Solid to Vapour	Sublimation	Sublimation	Deposition	Sublimation point

a.The phase change of the refrigerant in a refrigeration system is always between liquid and vapour

Table 7 Latent Heat Types and Characteristics

Each substance has its own latent heat characteristics, depending on the saturated temperature and pressure at which it is vaporised or condensed. Figure 24 illustrates the phase change processes of water with different amounts of latent heat at different saturated temperatures and pressures.

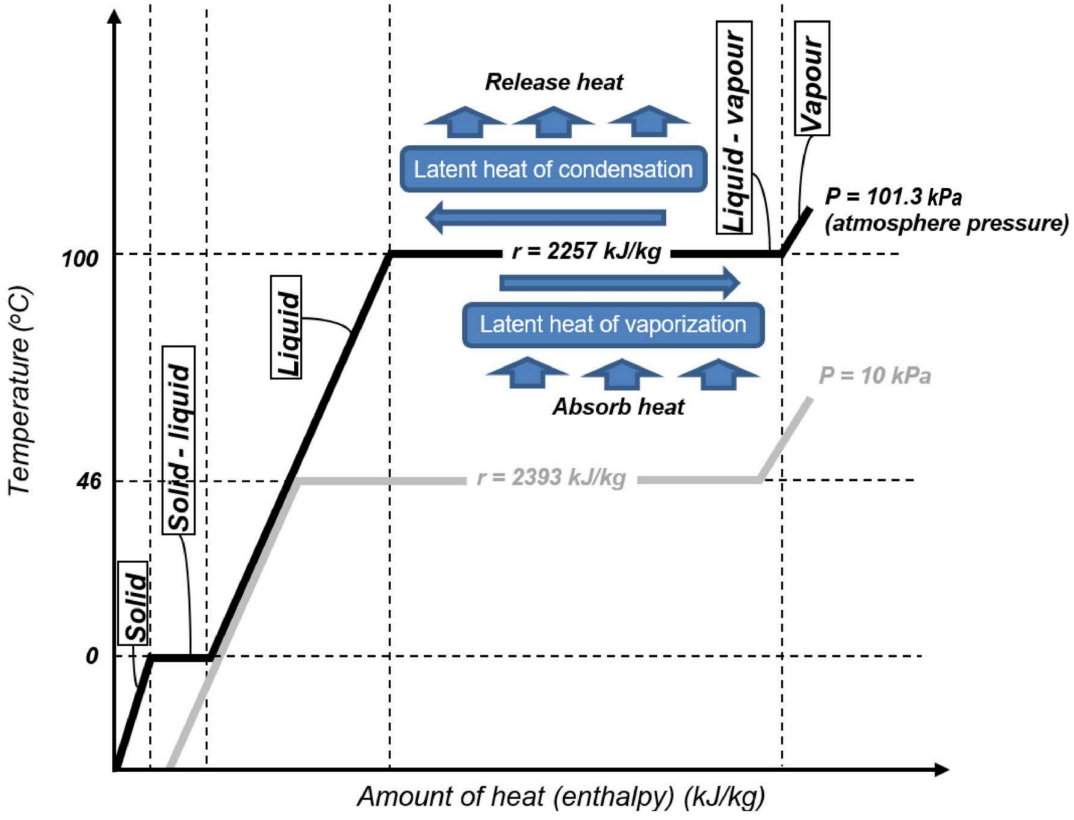


Figure 24 Phase Change and Latent Heat of Water

Water undergoes vaporization and condensation at 100°C at the atmosphere pressure of 101.3 kPa. It absorbs the latent heat of vaporization of 2257 kJ/kg. Similarly, the same amount of latent heat is given up during the condensation process at 100°C.

If the saturated pressure is lowered to 10 kPa, then the saturated temperature falls to 46°C. Under this condition, water absorbs the latent heat of vaporization of 2393 kJ/kg when it boils. If the water vapour is condensed at a pressure of 10 kPa, it will release the same latent heat of condensation (i.e. 2939 kJ/kg).

Refrigerant R134a (as used in the EC1500V refrigeration apparatus) vaporises at an atmospheric pressure of 101.3 kPa at a boiling point temperature of -26.1°C. Its latent heat of vaporisation is 216.8 kJ/kg. Figure 25 shows the latent heat variation of R134a under different saturated temperatures and pressures. Depending on the condensing or evaporating pressures of a refrigeration system, R134a will exhibit different amounts of latent heat.

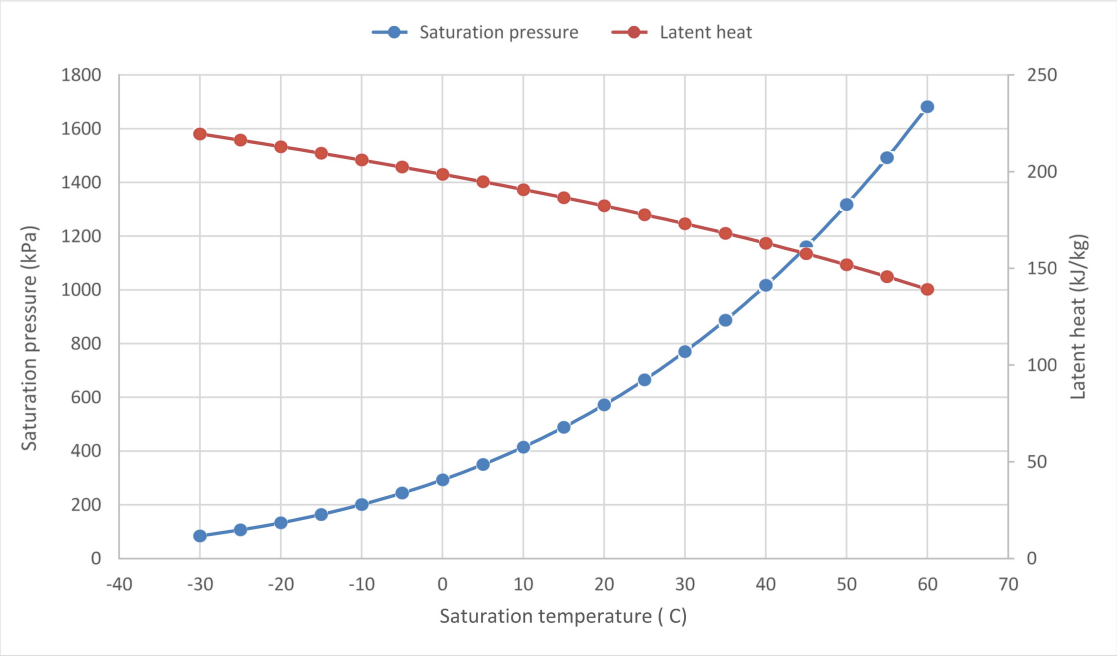


Figure 25 Graph showing the relationship between latent heat and the saturation pressure of R134a at variable saturation temperatures

Pressure – Enthalpy chart

It is usual to demonstrate a refrigeration cycle on a Pressure-Enthalpy chart to gain a better understanding of its thermodynamic processes as well as the different heat transfer processes occurring in the refrigeration system's components. The Pressure-Enthalpy chart shows values of absolute pressure on the vertical axis and specific enthalpy on the horizontal axis.

The state of the refrigerant at a given point on the diagram is defined by its properties including temperature, pressure, specific enthalpy, specific entropy, specific volume and quality (i.e. dryness). Each refrigerant has its individual Pressure-Enthalpy diagram. Figure 26 shows the Pressure-Enthalpy diagram for R134a. These charts often express terms in specific values so they are easily scalable for different mass flows.

Saturation Lines and Dryness

The (blue) curved line connecting the points where the refrigerant completely vaporises is called the saturated vapour line. The (blue) curved line connecting the points where the refrigerant condenses is called the saturated liquid line. The state of the refrigerant on the saturated vapour curve is saturated vapour with a dryness x=1. Similarly, the state of the refrigerant on the saturated liquid curve is saturated liquid with a dryness of x=0 (Figure 26).

The saturated liquid and vapour curves together form the dome shaped line with a so-called 'critical point' at the very top. To the left of the domed line the state of the refrigerant is liquid. To the right of the domed line the state of the refrigerant is vapour. The state of the refrigerant inside the domed area is a saturated liquid and vapour mixture (i.e. wet vapour). For R134a, the temperature and pressure are constant in this region.

The (blue) dryness lines (x) between the saturated liquid (x = 0) and vapour (x = 1) curve inside the dome represent the quality of the wet vapour. The larger value of x, the greater the ratio of vapour to liquid becomes.

Specific Enthalpy

Specific enthalpy (h), is defined as the heat content of the refrigerant per kg of refrigerant (kJ/kg). According to the first law of thermodynamics, it is the sum of specific internal energy "u" and specific flow energy or flow work "$p.V$". The horizontal axis of the chart shows values of specific enthalpy. It is often referred to as simply enthalpy.

$$h = u + p.V \qquad (1)$$

Pressure

Pressure (P), is denoted by an absolute value as shown in the vertical axes of the chart (Bar on the left axis and Pascal on the right) (Figure 26).

Gauge and Absolute Pressures Explained

Absolute pressure is the pressure with respect to absolute zero (a vacuum), this is normally used in calculations.

Gauge pressures are pressure with respect to atmospheric pressure. Zero gauge pressure is actually atmospheric pressure. Gauge pressure is often used on mechanical gauges. Gauges usually show pressure in bar as it is an easy unit to work with. These must be converted to Pascals (or $N.m^{-2)}$ for any calculations.

$$1\,N.m^{-2} = 1Pa = 0.00001bar$$

The EC1500V gauges and transducers measure the gauge pressure, but calculations use absolute pressure. The following may be used to convert gauge pressure into absolute pressure.

$$P_{LOW_ABS} = P_{LOW} + P_{atm}$$

$$P_{HIGH_ABS} = P_{HIGH} + P_{atm}$$

If there is no available measurement of the local atmospheric pressure available, then a value of 1.01325 bar (101.325 kPa) may be used.

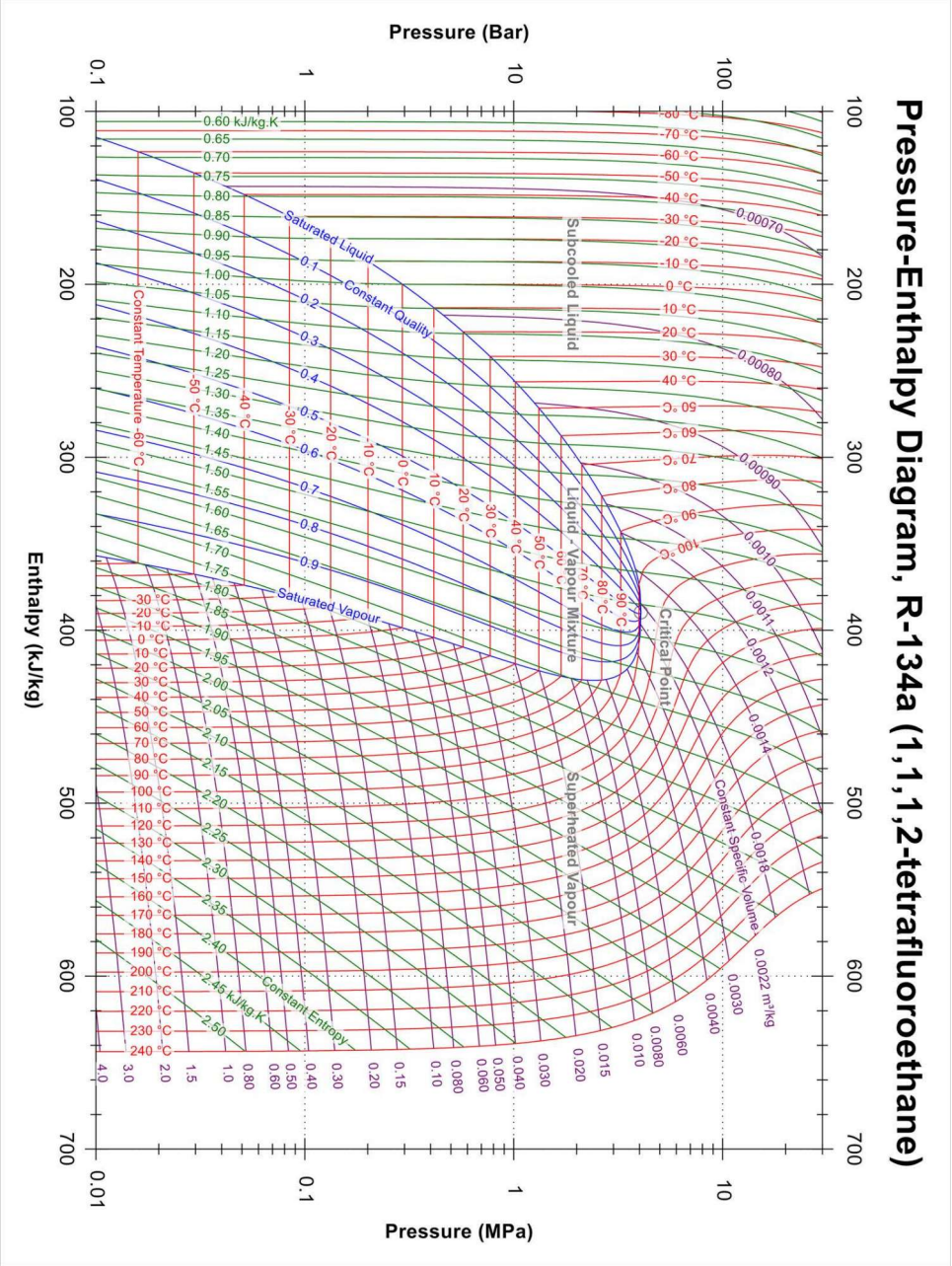


Figure 26 Pressure-Enthalpy diagram for R134a

NOTE *A PDF version of this chart can be found on the TecQuipment website at:*

https://www.tecquipment.com/downloads

Specific Entropy

Specific entropy (s), relates to the availability of energy, this indicates the amount of specific enthalpy able to be transferred per Kelvin (kJ/kg.K). As shown in the chart, the (green) entropy lines extend at an angle from the horizontal axis. The lower the enthalpy, the more vertical the entropy line becomes (Figure 26).

In an ideal refrigeration cycle these lines are relevant to isentropic compression. The difference between actual and isentropic compression processes is denoted by the isentropic efficiency of the compressor as follows.

$$\eta_s = \frac{h_{R2s} - h_{R1}}{h_{R2} - h_{R1}}$$

h_{R2s} is determined by following the constant isentropic line from the compressor suction state point 1 to the constant pressure line corresponding to discharge pressure at the state point 2, see **Figure 26** on page 34.

Specific Volume

Specific volume (v), is the amount of space (volume) that one kilogram of refrigerant occupies. The specific volume is represented by the (purple) upward slopping lines on the chart but flatter in slope than the entropy lines (Figure 26).

Temperature

Temperature (T), is denoted by (red) isotherm lines in the chart with a downward slope in the liquid and vapour regions but without a slope in the liquid-vapour mixture (dome) region. This mean that pressure and temperature are both constant in the mixture region (Figure 26).

Saturated Temperature

Saturated temperature (T_{sat}), is the temperature at which a vapour begins to condense (condensing temperature) or a liquid begins to vapourise (boiling temperature). At any point in the liquid/vapour mixture region (dome), the state of the refrigerant is at the saturation temperature.

Superheat

Superheat (SH), is the temperature difference between the vapour temperature and its saturated temperature at the same pressure. Superheat degree can be determined by subtracting suction temperature from the saturated vapour temperature at suction pressure.

$$SH = T_{R1} - T_{sat_suction} \qquad (2)$$

NOTE 

In EC1500V, the suction pressure is considered to be similar to low-side pressure $P_{LOW_ABS,}$ assuming that pressure loss is negligible in the suction-line pipe.

The SH degree is relevant to the operating conditions of the refrigeration system. If the SH degree is too high, the refrigeration system is not efficient. If the SH degree is too low, the compressor is put at a high risk of liquid flooding.

Discharge Superheat (DSH), this is different to suction superheat but is affected by suction superheat. To determine the discharge superheat degree, subtract the discharge temperature from the saturated vapour temperature at discharge pressure.

$$DSH = T_{R2} - T_{sat_discharge} \quad (3)$$

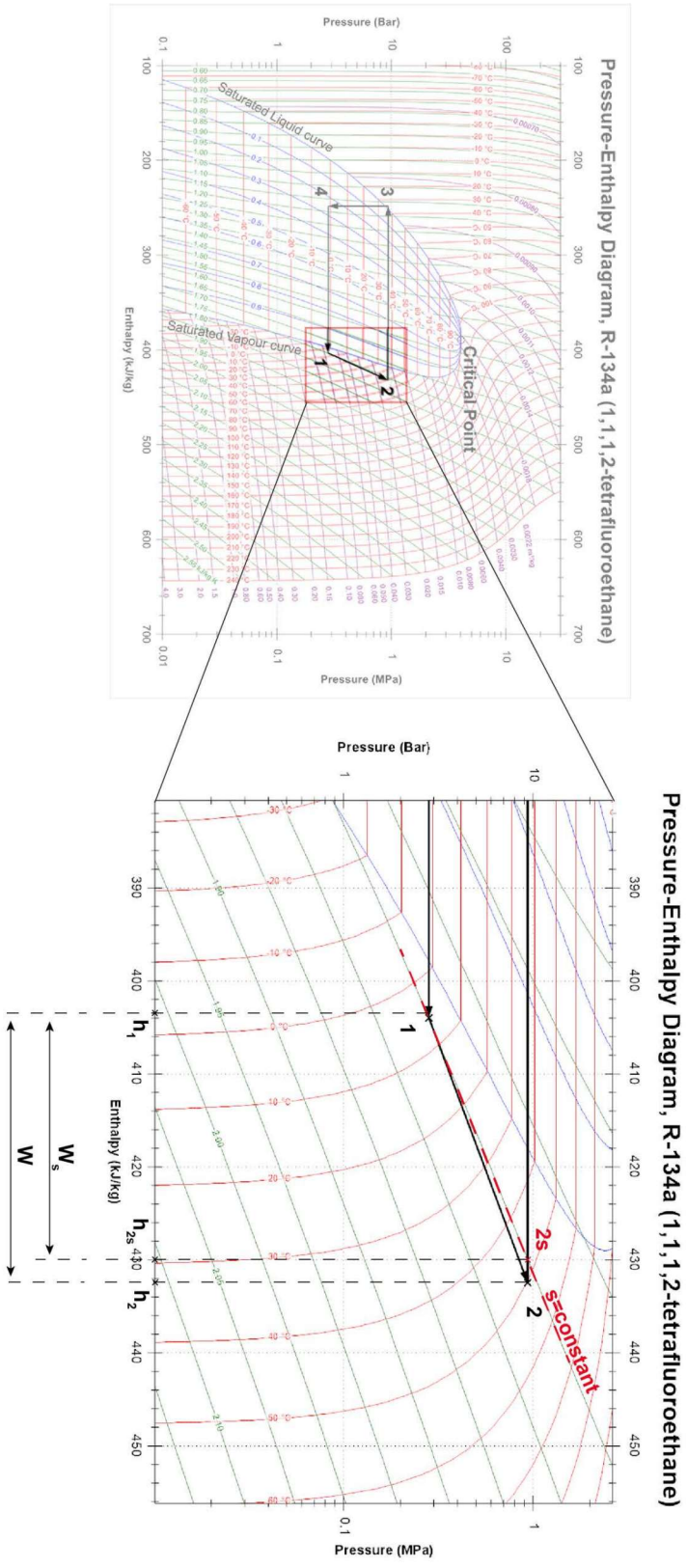


Figure 27 Actual and isentropic compression efficiencies on Pressure-Enthalpy chart

DSH is an indicator of the compressor operating conditions because it relates to discharge temperature. It should be maintained at a reasonable value as suggested by the compressor manufacturer.

NOTE 

In EC1500V, the discharge pressure is considered to be similar to high-side pressure $P_{HIGH_ABS,}$ assuming that pressure loss is negligible in the discharge-line pipe.

Subcooling

Subcooling (SC), is the temperature difference between the liquid temperature and its saturation temperature at the same pressure. The subcooling degree can be determined by subtracting the saturation liquid temperature from the liquid temperature at the TEV inlet at condensing pressure.

$$SC = T_{sat_condensation} - T_{R3} \qquad (4)$$

NOTE 

In EC1500V, the condensation pressure is considered to be similar to high-side pressure P_{HIGH_ABS}

Sensible Heat

Sensible heat (q), is the heat that is absorbed (superheating process) or released (desuperheating or subcooling processes) resulting in a change of temperature of the refrigerant.

Latent Heat

Latent heat (q), Refer to page 31 for details of vaporisation and condensation.

In practice, the heat exchange process in the condenser of a refrigeration system involves both sensible heat and latent heat as follows:

1. Removal of sensible heat to reduce the superheated vapour temperature to the saturated temperature.
2. Removal of the latent heat of condensation at the constant condensing temperature.
3. Removal of sensible heat to bring the saturated temperature down more by subcooling the liquid refrigerant.

The heat exchange process in the evaporator of a refrigeration system is accomplished by both:

1. The absorption of the latent heat of vaporisation at constant evaporating temperature.
2. The absorption of sensible heat to increase the saturated temperature by superheating the vapour refrigerant.

Main Components of a Vapour Compression Refrigeration System

Compressor

The compressor is used to circulate refrigerant through the system by increasing the pressure of the vapour in order to create a pressure difference between the condenser and the evaporator. The performance of a compressor is normally defined by its isentropic efficiency and volumetric efficiency.

Condenser

The condenser is a heat exchanger where refrigerant releases its sensible and latent heat to change phase from vapour to liquid. The performance of a condenser is defined by its heat exchange efficiency between the refrigerant and the heat sink. That is the ratio of the amount of heat received by the heat sink (actual heating effect) to the amount of heat rejected from the refrigerant (refrigerant heating effect).

Evaporator

The evaporator is also a heat exchanger, here the refrigerant absorbs sensible and latent heat to change phase from liquid to vapour. The performance of an evaporator in refrigeration is characterised by the ratio of the amount of heat absorbed by the refrigerant to the amount of heat received from the heat source.

Expansion Valve

The expansion valve is a refrigerant flow control device for reducing the pressure of the liquid refrigerant to equal the evaporating pressure. When the pressure drops, the temperature of the refrigerant also decreases to below that of the heat source.

Refrigerant

This is the working fluid of the refrigeration system. It transports heat from the heat source to the heat sink. Different types of refrigerant are used depending on the pressure required for the system and the temperature that needs to be obtained during refrigeration.

Thermodynamic processes of a simple vapour compression cycle

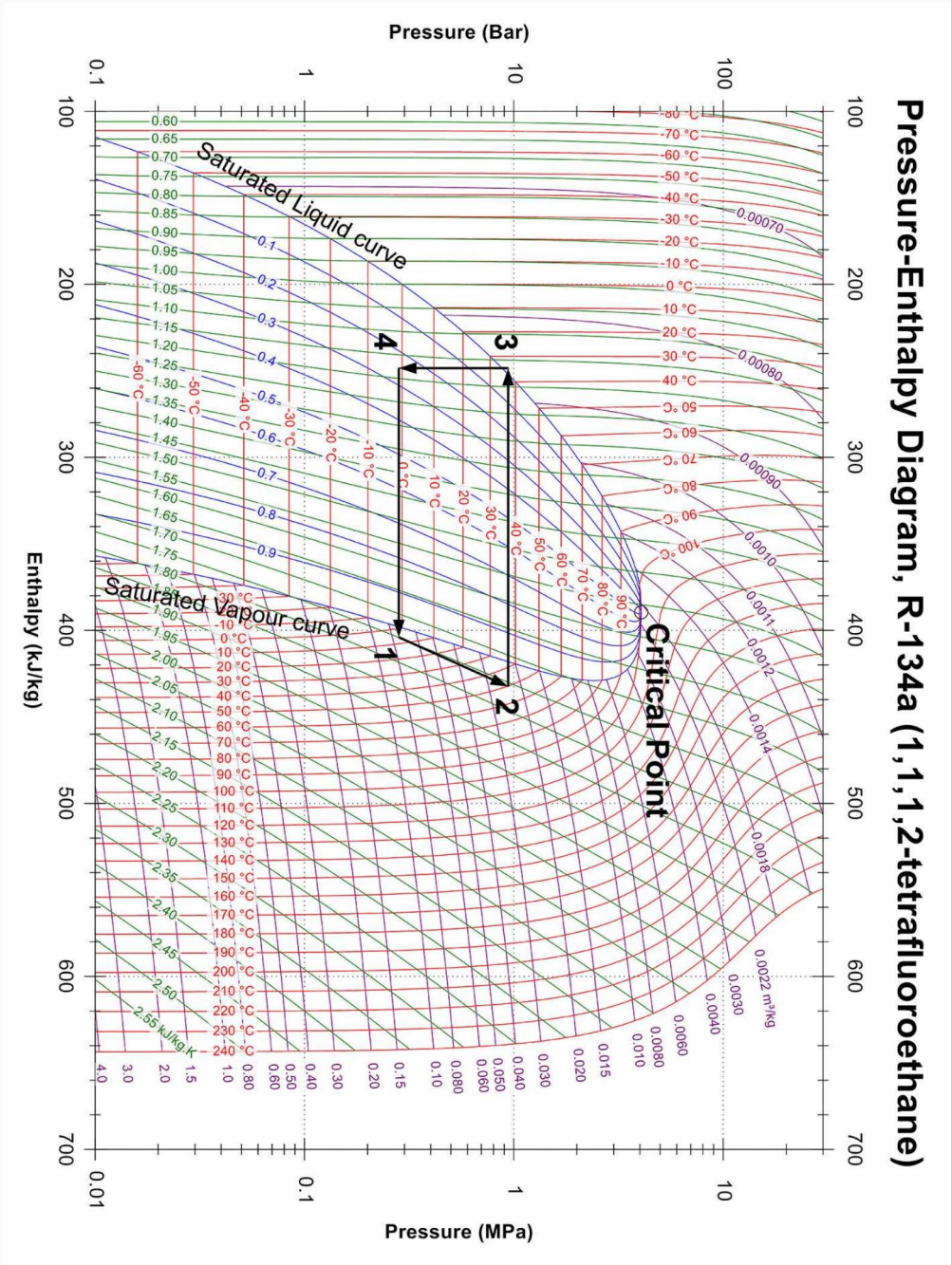


Figure 28 Pressure-Enthalpy diagram for R134a showing Refrigeration Cycle

NOTE *A PDF version of this chart can be found on the TecQuipment website at:*

https://www.tecquipment.com/downloads

The details of the refrigeration cycle as shown on the Pressure-Enthalpy chart (Figure 28) are as follows:

Process 1-2 Compression

The refrigerant enters the compressor as a superheated vapour at state 1 and is compressed to raise its temperature and pressure up to state 2.

Process 2-3 Condensation

After leaving the compressor, the superheated refrigerant vapour is passed through the condenser where it is de-superheated to release sensible heat. It is then condensed into liquid refrigerant to remove its latent heat at a constant pressure and temperature. It continues to release more heat to become a subcooled refrigerant liquid at state 3.

Process 3-4 Expansion

The sub-cooled refrigerant liquid is then expanded in the thermostatic expansion valve (TEV) to reduce its pressure, resulting in a decrease of its temperature. This forms a mixture of cold vapour-liquid refrigerant at the TEV outlet at state 4.

Process 4-1 Evaporation

The cold vapour-liquid refrigerant then moves to the evaporator in which it absorbs heat from the air in the duct causing the refrigerant to vaporise. This results in the air being cooled and a change in refrigerant state from a vapour-liquid mixture into a pure vapour. The refrigerant vapour is then superheated before returning to the compressor at state 1 to complete a basic refrigeration cycle.

State Changes of the Refrigerant in a Vapour Compression Refrigeration Cycle

The refrigerant is continuously changing its phase as it passes through the condenser and evaporation and expansion valves (Figure 29).

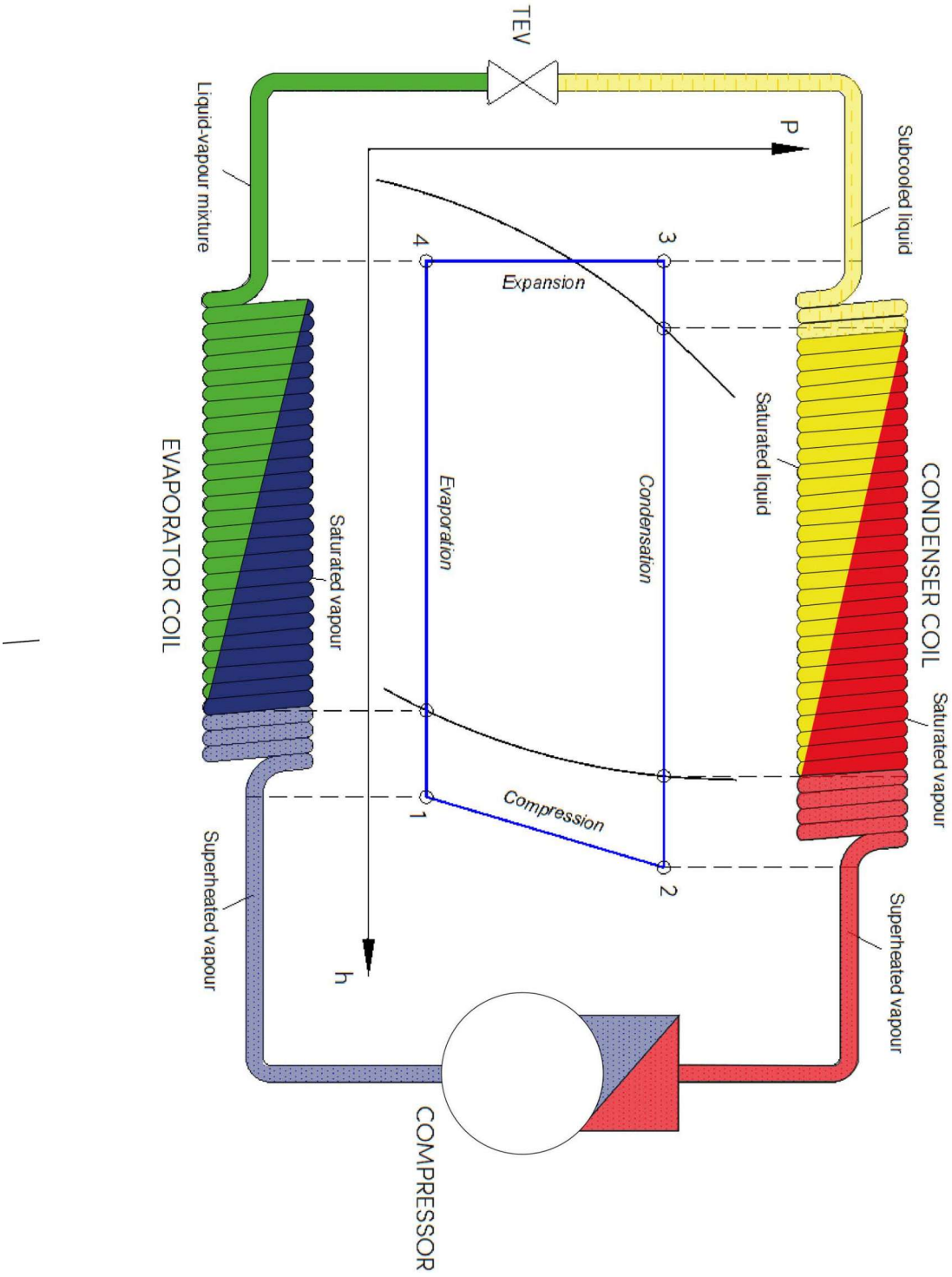


Figure 29 Phase Changes of the Refrigerant in a Refrigeration System

In the condenser, the refrigerant is de-superheated to become saturated vapour and then rejects its latent heat to change phase into a saturated liquid before being cooled further to attain a state of subcooled liquid.

In the TEV, the subcooled liquid is expanded to become a liquid-vapour mixture feeding to the evaporator.

In the evaporator, the liquid-vapour mixture is heated by the heat source to become a saturated vapour and subsequently the superheated vapour before reaching the compressor.

Performance analysis of the Refrigeration Cycle using the Pressure-Enthalpy Chart

Compressor

- Actual work of compression,

$$w_c = h_{R2} - h_{R1} \qquad (5)$$

This is the work input into the compressor shaft in order to compress the refrigerant vapour from a low pressure at state 1 to a higher pressure at state 2.

- Isentropic work of compression,

$$w_s = h_{R2s} - h_{R1} \qquad (6)$$

This is the work required for isentropic compression (i.e. q=0) to compress the refrigerant vapour from a low pressure at state 1 to a higher pressure at state *2s*. This is also the work put into the compressor in an ideal refrigeration cycle.

- Isentropic efficiency of compressor,

$$\eta_s = \frac{w_s}{w_c} = \frac{h_{R2s} - h_{R1}}{h_{R2} - h_{R1}} \qquad (7)$$

This is the ratio of isentropic work to actual work of the compressor.

- Pressure ratio

$$P_{ratio} = \frac{P_{HIGH_ABS}}{P_{LOW_ABS}}$$

This is the refrigeration compression ratio of the compressor

- Volumetric efficiency of compressor,

$$\eta_v = 0.97 - C\left[\left(\frac{1}{\left(\frac{Z_2}{Z_1}\right)}\right)\left(\frac{P_{HIGH_ABS}}{P_{LOW_ABS}}\right)^{\frac{1}{n}} - 1\right] \quad (8)$$

Where; Z_1 and Z_2 are compressibility factors at suction and discharge pressure conditions. These are determined from the generalized compressibility chart (these can be found in relevant publications, for example, see Introduction to Thermodynamics and Heat Transfer on page 77) in association with reduced temperature T_r and pressure P_r.

$$T_{r1} = \frac{T_{R1}}{T_c} \; ; \; T_{r2} = \frac{T_{R2}}{T_c}$$

$$P_{r1} = \frac{P_{LOW_ABS}}{P_c} \; ; \; P_{r2} = \frac{P_{HIGH_ABS}}{P_c}$$

Where; T_c and P_c are the critical temperature and pressure of R134a where:

T_c = 374.23 Kand P_c = 4070 kPa

Condenser

- Heating effect,

$$q_h = h_{R2} - h_{R3} \quad (9)$$

This is the amount of heat rejected to the surroundings per kg of refrigerant in the condenser due to the enthalpy change between states 2 and 3.

Expansion valve:

- Expansion of refrigerant at constant enthalpy,

$$h_{R3} = h_{R4} \quad (10)$$

This is an isenthalpic regulating process with no exchange of heat in order to bring the pressure down from a high pressure at state 3 to a lower pressure at state 4.

Evaporator:

- Refrigerating effect

$$q_c = h_{R1} - h_{R4} \quad (11)$$

This is the amount of heat absorbed from the air in the duct per kg of refrigerant in the evaporator due to the enthalpy change from state 4 to state 1

Coefficient of Performance (COP)

- Cooling Coefficient of Performance

$$COP_c = \frac{q_c}{w} = \frac{h_{R1} - h_{R4}}{h_{R2} - h_{R1}} \quad (12)$$

This is the ratio of the refrigerating effect to the actual work input to the compressor. It represents the efficiency of a refrigerator.

- Heating Coefficient of Performance

$$COP_h = \frac{q_h}{w} = \frac{h_{R2} - h_{R3}}{h_{R2} - h_{R1}} \quad (13)$$

This is the ratio of the heating effect to the actual work input into the compressor. It represents the efficiency of a heat pump.

Subcooling degree

$$SC = T_{sat_condensation} - T_3 \quad (14)$$

This is the difference between the refrigerant temperature at the condenser outlet and the saturated temperature of the refrigerant at condensation pressure.

Superheat degree

$$SH = T_1 - T_{sat_suction} \quad (15)$$

This is the difference between the refrigerant temperature at the compressor inlet and the saturated temperature of the refrigerant at suction pressure.

Vapour Compression Refrigeration Versus Reversed Carnot Cycles

The reversed Carnot cycle represents the most efficient refrigeration system. It is an ideal refrigeration cycle consisting of two isentropic (s=const) and two isothermal (T=const) processes as illustrated on the T- s chart in Figure 30 (the red cycle ABCD).

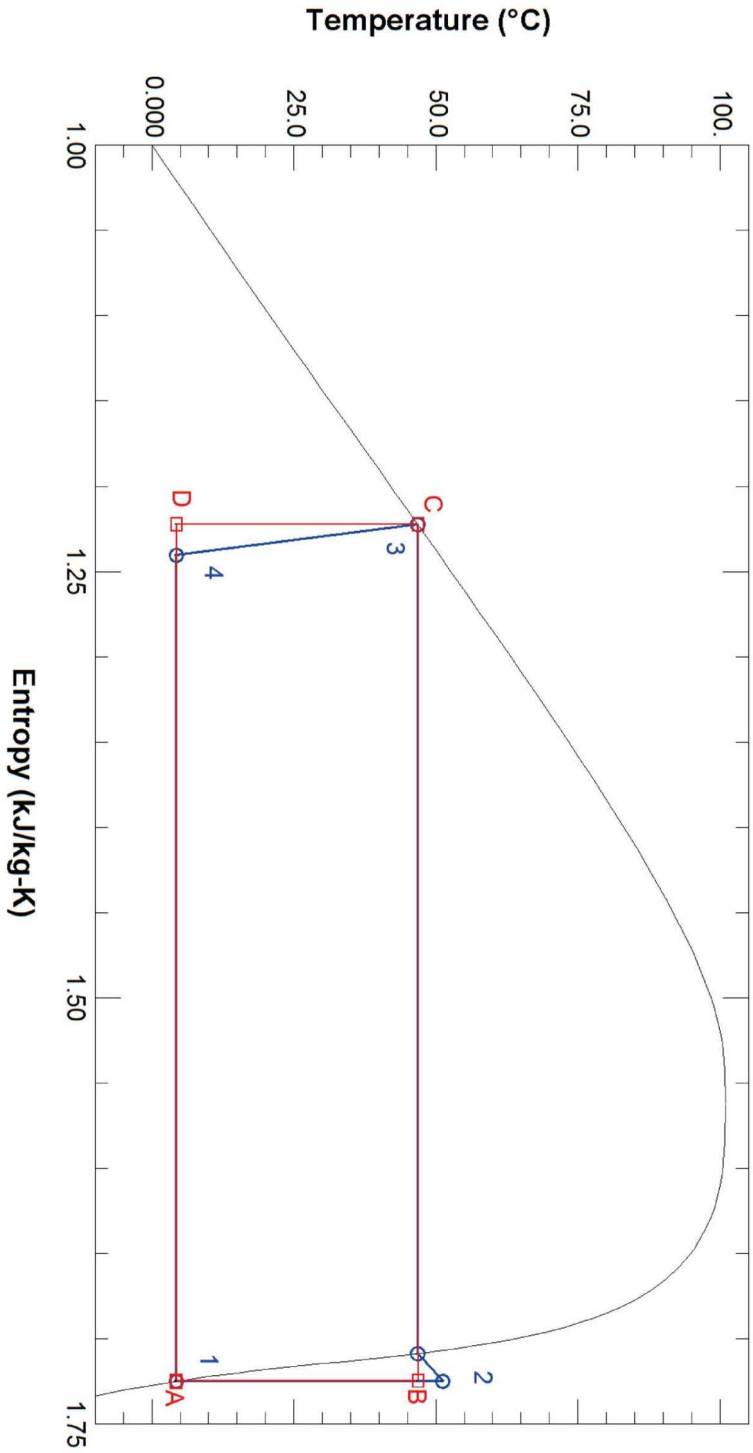


Figure 30 Comparison between Vapour Compression Refrigeration and the Carnot Cycles

In a Carnot cycle, the working fluid is compressed and expanded isentropically from process A to B and from process C to D, respectively (Figure 30). The heat rejection and absorption processes occur isothermally (from process B to C and process D to A respectively). The refrigerating and heating effects received from this ideal Carnot cycle are as follows:

$$q_{c_carnot} = T_A(s_A - s_D) \tag{16}$$

$$q_{h_carnot} = T_B(s_B - s_C) = T_B(s_A - s_D)$$

From the first law of thermodynamics, the compression work input is determined by the relationship:

$$w_{carnot} = q_{h_carnot} - q_{c_carnot} \tag{17}$$

$$= (T_B - T_A)(s_A - s_D)$$

Thus, the coefficient of performance of a reversed Carnot cycle is:

$$COP_{c_carnot} = \frac{q_{c_carnot}}{w_{carnot}} = \frac{T_A}{T_B - T_A} \tag{18}$$

$$COP_{h_carnot} = \frac{q_{h_carnot}}{w_{carnot}} = \frac{T_B}{T_B - T_A} \tag{19}$$

In conclusion, the coefficient of performance of a reversed Carnot cycle is dependent only on heat sink and heat source temperatures.

Experiments

Safety

WARNING 

The pressure switch is factory set at default values as follows:

Low pressure side: Pressure 1 bar, Differential 1 bar

High pressure side: Pressure 14 bar, Differential 4 bar fixed.

Changing these default settings may result in potential risks and the incorrect operation of the apparatus.

WARNING 

If either of the pressure warning lamps are illuminated, or the unit is damaged in any way, do not attempt to repair or re-start it. Please isolate the apparatus and contact TecQuipment or local agent.

WARNING 

Take care not to splash the water onto the equipment. Always clear up water spills immediately.

CAUTION 

Always replace the dust cap to the VDAS® socket when not in use to prevent damage from dust or water entering.

CAUTION 

Always leave the compressor for at least five minutes after it has been switched off before it is switched on again, or it could be damaged.

CAUTION 

The refrigeration system is supplied charged with the correct amount of refrigerant. In the case of a suspected refrigerant leak, please contact a certified service engineer to fix the problem before attempting to carry out any experiments.

General

Base Line

Each experiment must start from the same base point as follows:

- The water in both tanks must be between 20 and 23°C.
- The water in both tanks should be at a similar temperature ($T_{W1} \sim T_{W2}$).

Therefore if a previous experiment has just been run, the water must be changed.

Experiments 2, 3 and 5

If a previous experiment has not been performed immediately before experiments 2, 3 or 5 then the system must be run to a steady state first (as instructed in the experiment details).

The water must then be drained and replace by fresh water between 20 and 23°C, before the new experiment can take place.

Water

Each experiment needs 7 litres of water. If several experiments are to be run one after the other then 7 litres of water will be required for each experiment e.g. if three experiments are to be run at least 21 litres of (3 x 7 litres) of water will be required.

Experiment 1 – Demonstration and performance analysis of a vapour compression refrigeration or heat pump cycle

Aim

To learn the thermodynamic processes of a vapour compression refrigeration or heat pump cycle and demonstrate it on a Pressure-Enthalpy chart.

Preparation

1. About 7 litres clean water (preferably de-mineralised or de-ionised) at about 20-23°C
2. Pressure-Enthalpy chart for R134a
3. Pencil
4. Ruler
5. A suitable cloth to clean up water spills
6. Instead of items 2 - 4 TecQuipment's integrated VDAS® can be used with a suitable computer.

Procedure

1. Fill the two water tanks up to the lower surface of the water tank's top plate. See **Figure 18** on page 21. (Fresh water must be used if the system has just been used for another experiment).
2. If not using VDAS®, create a blank result table similar to Table 8 below.
3. Make sure the unit is plugged in and the main switch on the left side of the control panel is ON.
4. Turn the water pump ON using the pump switch on the control panel.

CAUTION 

Once the pump is switched on, the water level in the left hand tank will rise a little and the level in the right hand tank will drop.

Top the right tank back up to the lower surface of its top plate (Figure 18)

5. Wait until the water temperature between the two tanks is similar ($T_{W1} \sim T_{W2}$).
6. Turn the compressor ON using the compressor switch on the control panel.
7. Leave the unit to run until the **temperature difference** between the two tanks is stable.
8. If not using VDAS® record the measured values as shown in Table 8.
9. Switch OFF the compressor and the pump.
10. Drain the tanks See **Draining the Water** on page 25.

Parameter	Symbol	Experiment case with water pump “ON”
Compressor suction temperature (°C)	T_{R1}	
Compressor discharge temperature (°C)	T_{R2}	
TEV inlet temperature (°C)	T_{R3}	
TEV outlet temperature (°C)	T_{R4}	
Low-side pressure (bar)	P_{LOW}	
High-side pressure (bar)	P_{HIGH}	
Temperature difference between the two tanks BEFORE the compressor was switched “ON” (°C)	ΔT_W	
Temperature difference between the two tanks AFTER the compressor was switched ON (°C)	ΔT_W	

Table 8 Experiment 1 Table

NOTE  *If using the Tequipment supplied chart please note that the left y-axis shows “bar” and the right y axis shows “MPa”. Be sure to read from the correct axis when calculating your results.*

Results Analysis

1. Convert gauge pressures (P_{HIGH}, P_{LOW}) into values of absolute pressure (P_{HIGH_ABS}, P_{LOW_ABS}) as used in the Pressure-Enthalpy chart.

2. Plot the state points of the refrigerant on the Pressure-Enthalpy chart as follows:

   Point 1: use the measured values of T_{R1} and absolute pressure of the low-side (P_{LOW_ABS}).

   Point 2: use the measured values of T_{R2} and absolute pressure of the high-side (P_{HIGH_ABS}).

   Point 3: use the measured value of T_{R3} and the pressure of the condensation process (equivalent to the absolute pressure of the high-side (P_{HIGH_ABS})).

   Point 4: use the measured value of vaporisation pressure (equivalent to the absolute pressure of the low-side (P_{LOW_ABS})) and the constant enthalpy of the expansion process ($h_{R3} = h_{R4}$).

3. Draw lines to connect points 1-2, 2-3, 3-4 and 4-1 to achieve a complete refrigeration cycle.

4. Determine the cycle performance by calculating the work of compression, the heating effect, the refrigerating effect and the two heating and cooling COPs using formulas (6, 10, 12, 13, and 14) also the superheat and subcooling degrees using formulas (15 and 16).

5. How does the water temperature of the two tanks change after the compressor is switched ON? How does this affect the cycle performance?

Experiment 2 – Temperature-pressure relationship

Aim

To explore the relationship between the temperature and the pressure of R134a at saturation conditions.

Preparation

1. About 7 litres water (preferably de-mineralised or de-ionised) at about 20-23°C
2. Pressure-Enthalpy chart for R134a
3. Pencil
4. Ruler
5. A suitable cloth to clean up water spills
6. Instead of items 2 - 4 TecQuipment's integrated VDAS® can be used with a suitable computer.

Procedure

NOTE 

Another experiment should be run immediately before Experiment 2 to ensure that the system completes a suitable start-up process. If this is not the case please follow the procedure in step 2 below.

1. Make sure the unit is plugged in and the main switch is ON.
2. If experiment 1 has **not** been run immediately prior to starting this experiment, fill the two tanks with water at 20-23°C and run the pump and compressor for 15 minutes.
3. Once the previous experiment is finished. Switch OFF the compressor and the pump. Drain the tanks (see **Draining the Water** on page 25).
4. Turn the pump OFF once both tanks are completely drained.
5. Fill the two water tanks with fresh water at 20-23°C, up to the lower surface of the water tank's top plate. See **Figure 18** on page 21.
6. Switch the pump ON and run the system until both tanks are at a similar temperature: $T_{W1} \sim T_{W2}$.

CAUTION 

Once the pump is switched on, the water level in the right hand tank will drop.

Top the tank back up to the lower surface of its top plate (Figure 18)

7. Switch the pump OFF.
8. Switch the compressor ON
9. If not using VDAS®, create a blank result table similar to Table 9.

10. If not using VDAS® record the measured values as shown in Table 9 while the hot water tank is heating up.

NOTE  *The water pump is OFF from step 7 of this experiment.*

WARNING  ***Due to the maximum allowable working pressure of the refrigerant in the condenser coil, do not heat the left (hot water) tank to over 45°C.***

11. Switch OFF the compressor and drain the tanks See **Draining the Water** on page 25.

Parameter	Symbol	Hot Water Tank Tempearture T_{W1} (°C)			
		25	30	35	40
Gauge pressure	P_{GAUGE}				
Saturated temperature as shown on high pressure gauge	T_{GAUGE}				
High-side pressure	P_{HIGH}				
Saturated temperature as found from the Pressure-Enthalpy chart	T_{sat}				

Table 9 Experiment 2 Table

Results Analysis

1. Convert the high-side pressure P_{HIGH} into absolute pressure P_{HIGH_ABS} and use it to find the corresponding saturation temperatures T_{sat} from the Pressure-Enthalpy chart.

2. Compare the temperature as shown on the high pressure gauge T_{GAUGE} with the saturation temperature from the Pressure-Enthalpy chart T_{sat}.

3. Plot a graph to demonstrate a relationship between the absolute pressure P_{HIGH_ABS} and saturation temperature $T_{sat.}$

4. Repeat the experiment for the low-side pressure P_{LOW} and find the relationship between P_{LOW_ABS} and the saturation temperature T_{sat}. (This will take some time as the temperature of the cold water tank will drop very slowly).

5. Why does heat exchange occur between the refrigerant and the water while the saturated pressure and temperature of the refrigerant remain constant?

Experiment 3 – Performance of refrigeration or heat pump cycle under different heat source and heat sink temperatures

Aim

To evaluate the effect of different heat sink and heat source temperatures on cycle performance.

Preparation

1. About 7 litres water (preferably de-mineralised or de-ionised) at about 20-23°C
2. Pressure-Enthalpy chart for R134a
3. Pencil
4. Ruler
5. A suitable cloth to clean up water spills
6. Instead of items 2 - 4 TecQuipment's integrated VDAS® can be used with a suitable computer.

Procedure

NOTE 

An earlier experiment should be performed immediately before Experiment 3 to ensure that the system completes a suitable start up process. If this is not the case, please make sure that the system has been warmed up (see step 2 below).

1. Make sure the unit is plugged in and the main switch is ON.
2. If an earlier experiment has **not** been run immediately prior to starting this experiment, fill the two tanks with water at 20-23°C and run the pump and compressor for 15 minutes.
3. Once the system is at a steady state. Switch OFF the compressor and the pump. Drain the tanks (see **Draining the Water** on page 25).
4. Turn the pump OFF once both tanks are completely drained.
5. Fill the two water tanks with fresh water at 20-23°C, to the lower surface of the water tank's top plate. See **Figure 18** on page 21.
6. Switch the pump ON and run the system until both tanks are at a similar temperature: $T_1 \sim T_2$.

CAUTION 

Once the pump is switched on, the water level in the right hand tank will drop.

Top the tank back up to the lower surface of its top plate (Figure 18)

7. Switch the pump OFF
8. Switch the compressor ON.

9. If not using VDAS®, create a blank result table similar to Table 10.
10. If not using VDAS®, record the measured values as shown in Table 10 while the hot water tank is heating up.

WARNING  ***Due to the maximum allowable working pressure of the refrigerant in the condenser coil, do not heat the left (hot water) tank to over 45 °C.***

11. Switch OFF the compressor and drain the tanks See **Draining the Water** on page 25.

NOTE  *The water pump is OFF from step 7 of this experiment.*

Parameter	Symbol	Hot Water Tank Temperature T_{W1} (°C)		
		30	35	40
Cold water tank temperature	T_{W2}			
Low-side pressure	P_{LOW}			
High-side pressure	P_{HIGH}			
Suction-line temperature	T_{R1}			
Discharge-line temperature	T_{R2}			
TEV inlet temperature	T_{R3}			
TEV outlet temperature	T_{R4}			

Table 10 Experiment 3 Table

Results Analysis

1. Use a Pressure-Enthalpy chart to find the other properties of R134a including saturation temperature (T_{sat}) and specific enthalpy (h_{R1}, h_{R2}, h_{R3}) under measured temperatures and pressures of refrigerant.
2. Calculate the following:
   - Cycle performance as instructed in Experiment 1,
   - The water temperature difference between the hot and cold tanks ΔT_W,
   - The pressure ratio between the high-side and low-side pressures P_{ratio}.
3. Plot a graph to demonstrate the relationship between the COPs and pressure ratio for various water temperature differences.
4. How does the COP vary with water temperature difference and pressure ratio.

Experiment 4 – Determination of isentropic and volumetric efficiencies of the compressor

Aim

To evaluate compressor performance by determining isentropic and volumetric efficiencies.

Preparation

1. About 7 litres water (preferably de-mineralised or de-ionised) at about 20-23°C
2. Pressure-Enthalpy chart for R134a
3. Pencil
4. Ruler
5. A suitable cloth to clean up water spills
6. Instead of items 2 - 4 TecQuipment's integrated VDAS® can be used with a suitable computer.

Procedure

NOTE 

An earlier experiment should be performed immediately before Experiment 4 to ensure that the system completes a suitable start up process. If this is not the case, please make sure that the system has been warmed up (see step 2 below).

1. Make sure the unit is plugged in and the main switch is ON.
2. If an earlier experiment has **not** been run immediately prior to starting this experiment, fill the two tanks with water at 20-23°C and run the pump and compressor for 15 minutes.
3. Once the system is at a steady state. Switch OFF the compressor and the pump. Drain the tanks (see **Draining the Water** on page 25).
4. Turn the pump OFF once both tanks are completely drained.
5. Fill the two water tanks with fresh water at 20-23°C, to the lower surface of the water tank's top plate. See **Figure 18** on page 21.
6. Switch the pump ON and run the system until both tanks are at a similar temperature: $T_1 \sim T_2$.

CAUTION 

Once the pump is switched on, the water level in the right hand tank will drop.

Top the tank back up to the lower surface of its top plate (Figure 18)

7. Switch the compressor ON
8. Leave the unit to run until the **temperature difference** between the two tanks is stable.

9. Take the first data set.
10. Switch the pump OFF and leave the compressor running.
11. Take two more data sets at T_{W1} =35°C and again at T_{W1} =40°C.
12. If not using VDAS®, create a blank result table similar to Table 11.
13. Record the measured values as shown in Table 11 while the hot water tank is heating up.

WARNING  ***Due to the maximum allowable working pressure of the refrigerant in the condenser coil, do not heat the left (hot water) tank to over 45 °C.***

14. Switch the compressor OFF and drain the tanks See **Draining the Water** on page 25.

NOTE  *The water pump is OFF from step 10 of this experiment.*

		Pump ON	Pump OFF	
Parameter	**Symbol**	**T_{W1} = 30 °C**	**T_{W1} = 35 °C**	**T_{W1} = 40 °C**
Cold water tank temperature (°C)	T_{W2}			
Water temperature difference (°C)	ΔT_W			
Low-side pressure (bar)	P_{LOW}			
High-side pressure (bar)	P_{HIGH}			
Suction-line temperature (°C)	T_{R1}			
Discharge-Line temperature (°C)	T_{R2}			

Table 11 Experiment 4 Table

Results Analysis

1. Use the Pressure-Enthalpy chart to find the other properties of R134a including specific enthalpy (h_{R1}, h_{R2}) and specific entropy (s_{R1}, s_{R2}) under measured values of refrigerant temperature and pressure.

2. Calculate the work input of actual compression process using formula (6).

3. Use the Pressure-Enthalpy chart to find the enthalpy of the final state point (h_{2s}) of an isentropic compression. Do this by drawing an entropy line passing over the initial state point (T_{R1}, P_{LOW_ABS}) to cut the pressure line at P_{HIGH_ABS}. Read the value of specific entropy (h_{2s}).

4. Calculate the work input of the isentropic compression process using formula (7).

5. Determine the isentropic efficiency of the compressor using formula (8).

6. Repeat the analysis of isentropic efficiencies for various water temperature differences between the hot and cold tanks.

7. Plot a graph to demonstrate the variation of these efficiencies with pressure ratio.

Further Analysis

1. Using the generalised compressibility charts from relevant literature (e.g. see Introduction to Thermodynamics and Heat Transfer on page 77), determine the volumetric efficiency of the compressor using formula (9). Using a value of percent clearance C = 10% for the compressor and a polytropic exponent n = 1.35 to calculate volumetric efficiency.

NOTE  *You may assume a compressibility factor ratio* $\left(\frac{Z_2}{Z_1}\right)$ *of 0.9 if additional sources are not available.*

Experiment 5 – Performance comparison between actual vapour compression refrigeration and reversed Carnot cycle

Aim

To compare performance between an actual refrigeration or heat pump cycle with the reversed Carnot cycle (i.e. an ideal refrigeration or heat pump cycle)

Preparation

1. About 7 litres water (preferably de-mineralised or de-ionised) at about 20-23°C
2. Pressure-Enthalpy chart for R134a
3. Pencil
4. Ruler
5. A suitable cloth to clean up water spills
6. Instead of items 2 - 4 TecQuipment's integrated VDAS® can be used with a suitable computer.

Procedure

NOTE 

Another experiment should be run immediately before Experiment 5 to ensure that the system completes a suitable start up process. If this is not the case please follow the procedure in step 2 below.

1. Make sure the unit is plugged in and the main switch is ON.
2. If another experiment has **not** been run immediately prior to starting this experiment, fill the two tanks with water at 20-23°C and run the pump and compressor for 15 minutes.
3. Once the previous experiment is finished. Switch the compressor OFF and the pump. Drain the tanks (see **Draining the Water** on page 25).
4. Turn the pump OFF once both tanks are completely drained.
5. Fill the two water tanks with fresh water at 20-23°C, up to the lower surface of the water tank's top plate. See **Figure 18** on page 21.
6. Switch the pump ON and run the system until both tanks are at a similar temperature: $T_{W1} \sim T_{W2}$.

CAUTION 

Once the pump is switched on, the water level in the right hand tank will drop.

Top the tank back up to the lower surface of its top plate (Figure 18)

7. Switch the pump OFF.

8. Switch the compressor ON.
9. If not using VDAS®, create a blank result table similar to Table 12.
10. Record the measured values as shown in Table 12.

WARNING  ***Due to the maximum allowable working pressure of the refrigerant in the condenser coil, do not heat the left (hot water) tank to over 45 °C.***

11. Switch the compressor OFF and drain the tanks See **Draining the Water** on page 25.

NOTE  *The water pump is OFF from step 7 of this experiment.*

Parameter	Symbol	Hot Water Tank Temperature T_{W1} = 40°C
Cold water tank temperature	T_{W2}	
Low-side pressure	P_{LOW}	
High-side pressure	P_{HIGH}	
Suction-line temperature	T_{R1}	
Discharge-line temperature	T_{R2}	
TEV inlet temperature	T_{R3}	
TEV outlet temperature	T_{R4}	

Table 12 Experiment 5 Table

Results Analysis

1. Calculate the heating and cooling COPs of the refrigeration cycle as instructed in experiment 1.
2. Calculate the heating and cooling COPs of the reversed Carnot cycle using formulas (19) and (20) with heat source temperature $T_A \sim T_{W2}$ and heat sink temperature $T_B \sim T_{W1}$.
3. Compare COPs between the actual refrigeration cycle and the reversed Carnot cycle.